

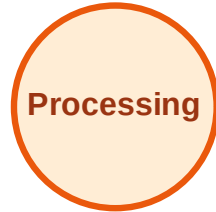
BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing



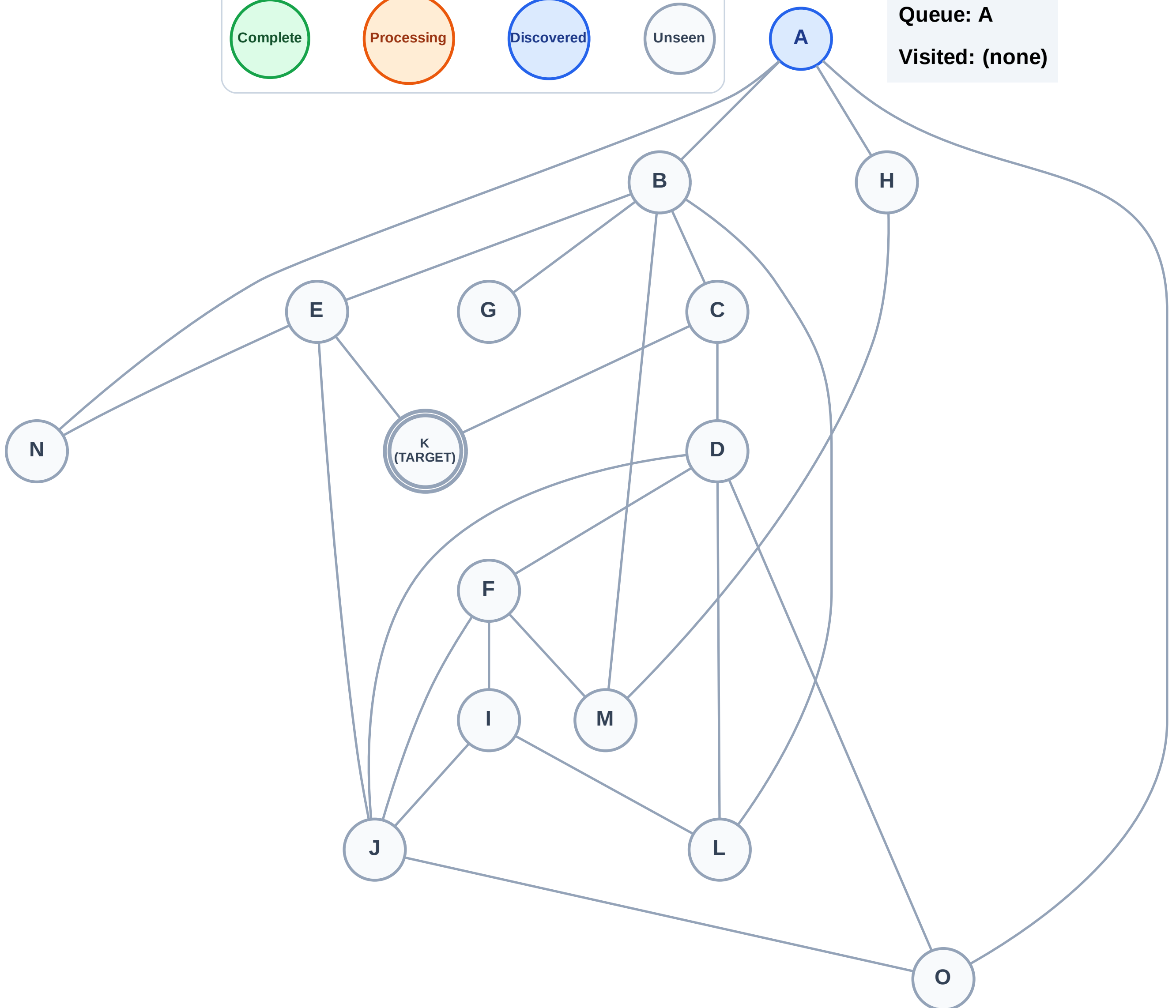
Discovered



Unseen

Queue: A

Visited: (none)



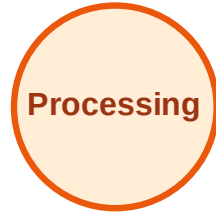
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



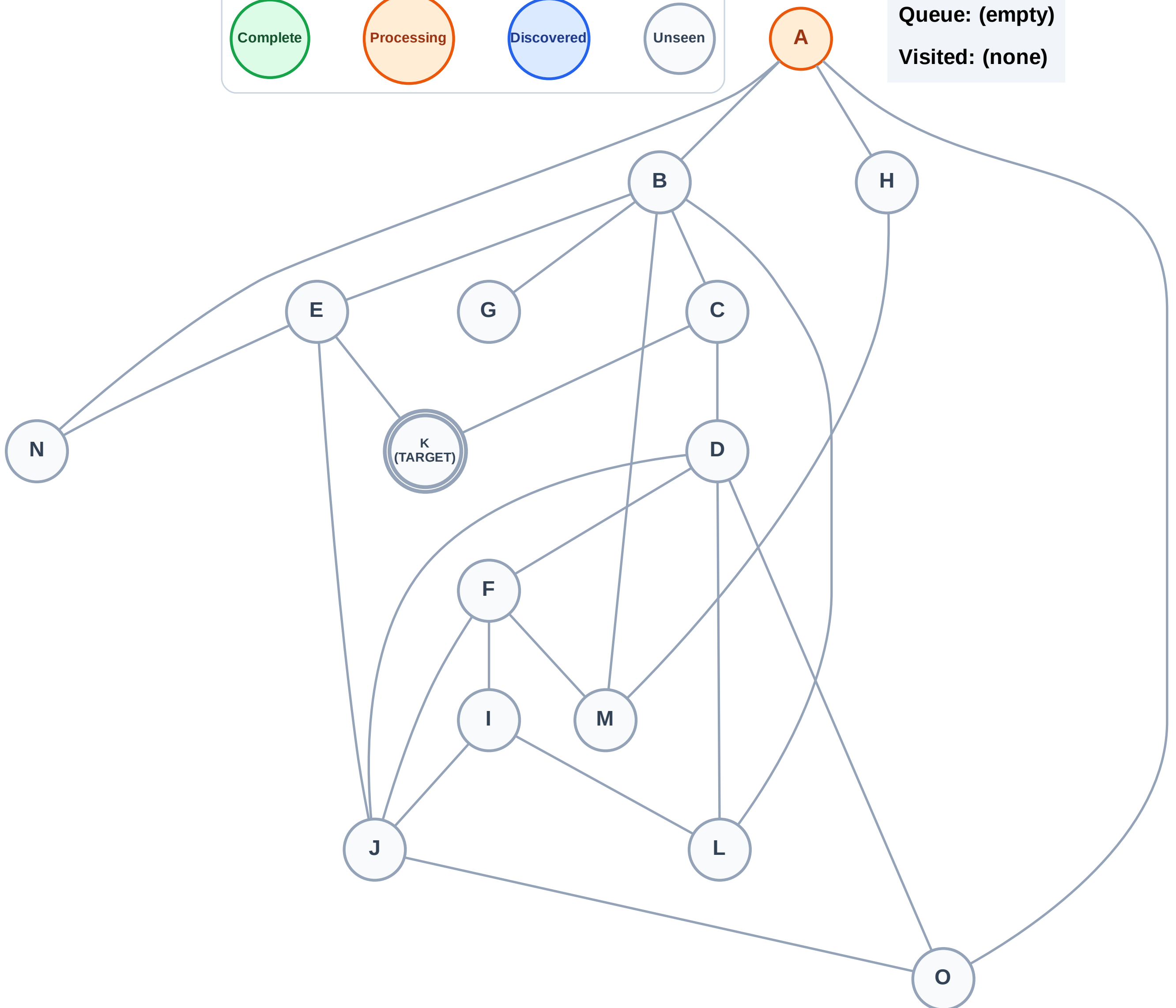
Discovered



Unseen

Queue: (empty)

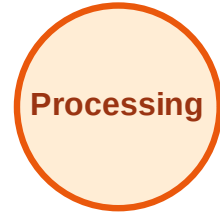
Visited: (none)



BFS Traversal

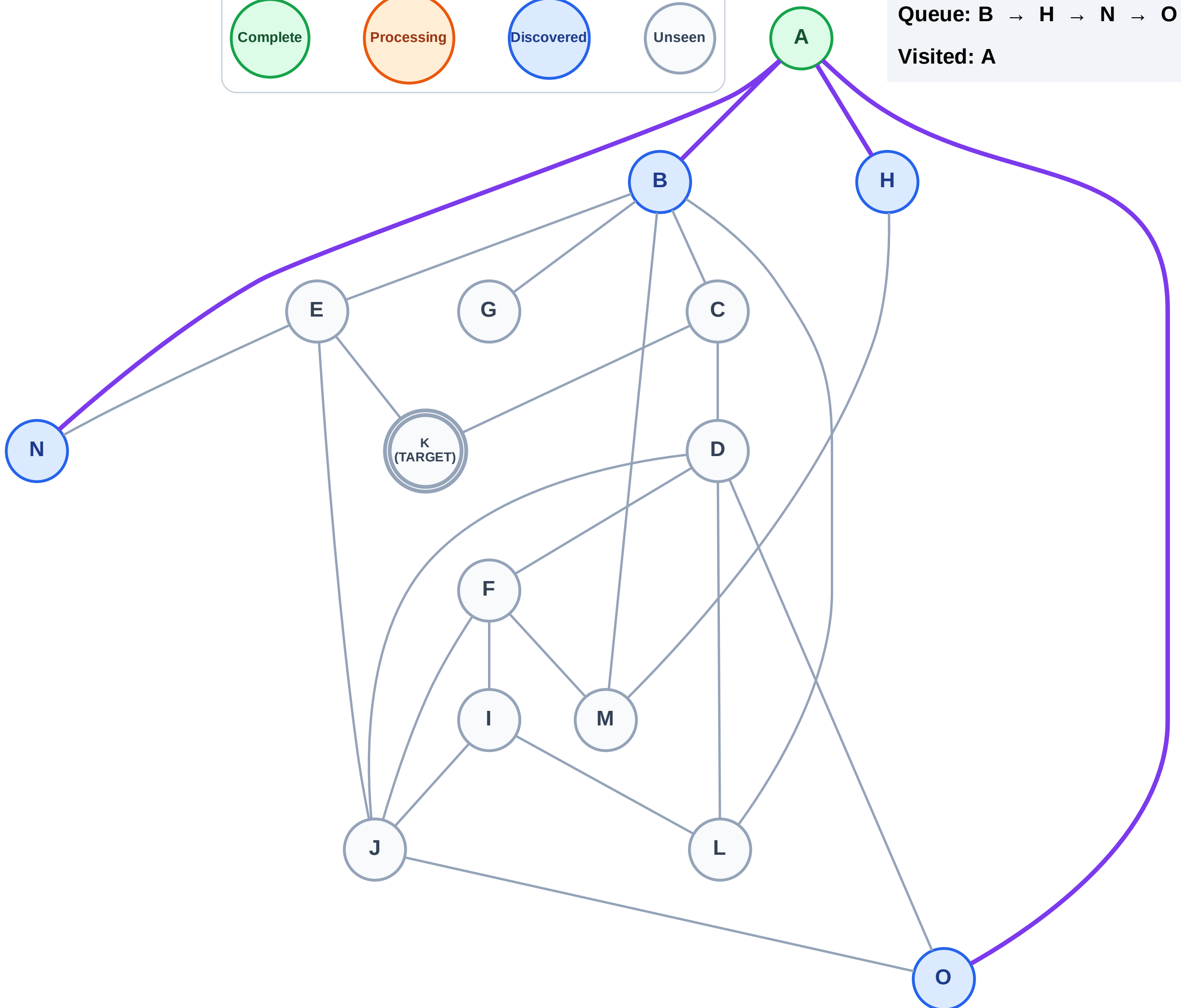
Step 3: Discover B, H, N, O from A

Legend



Queue: B → H → N → O

Visited: A



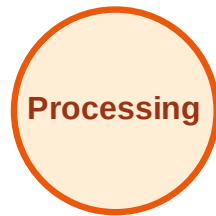
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



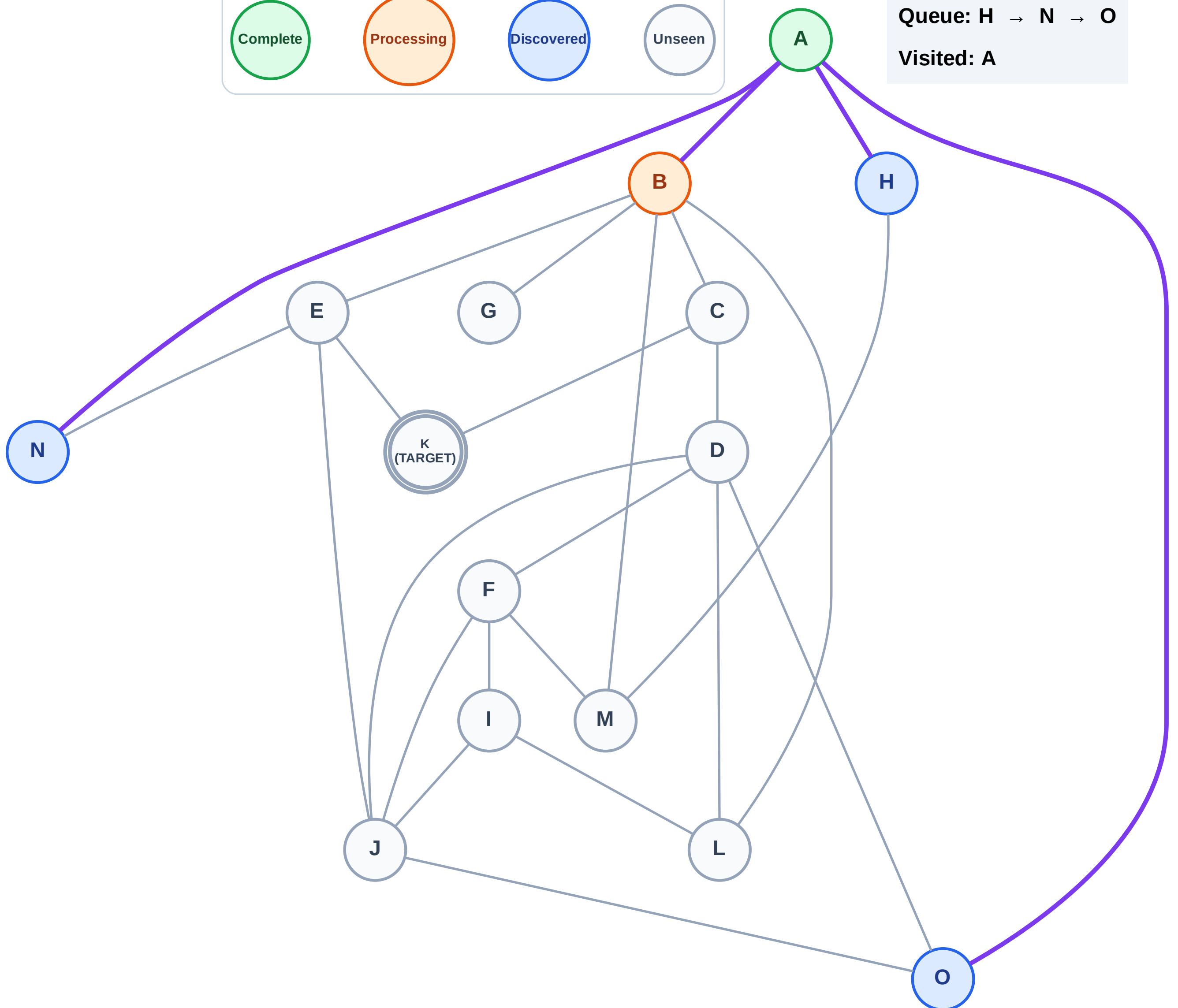
Discovered



Unseen

Queue: H → N → O

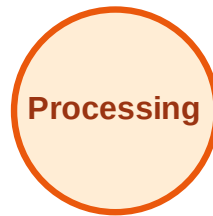
Visited: A



BFS Traversal

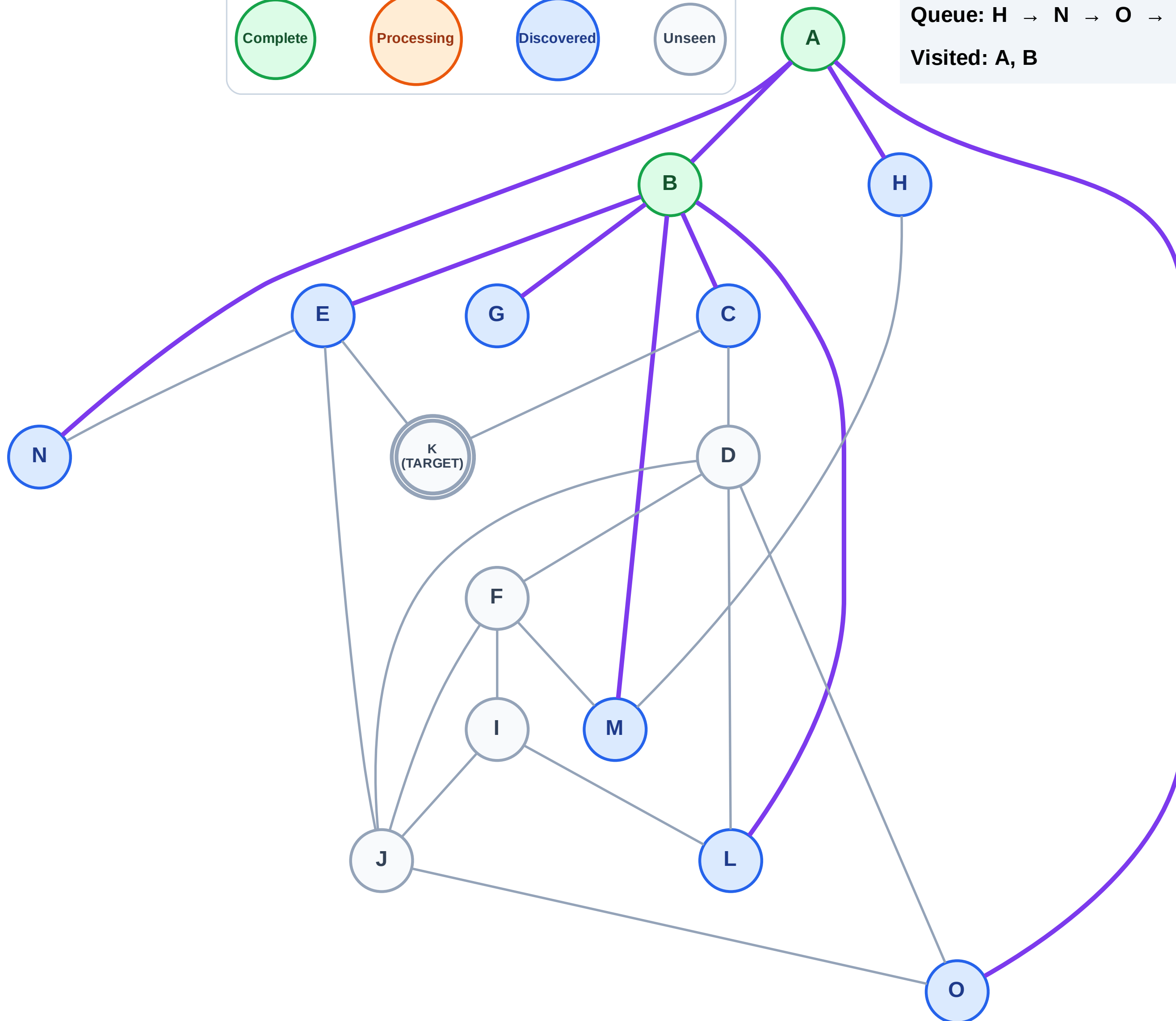
Step 5: Discover C, E, G, L, M from B

Legend



Queue: H → N → O → C → E → G → L → M

Visited: A, B



BFS Traversal

Step 6: Process node H

Legend

Complete

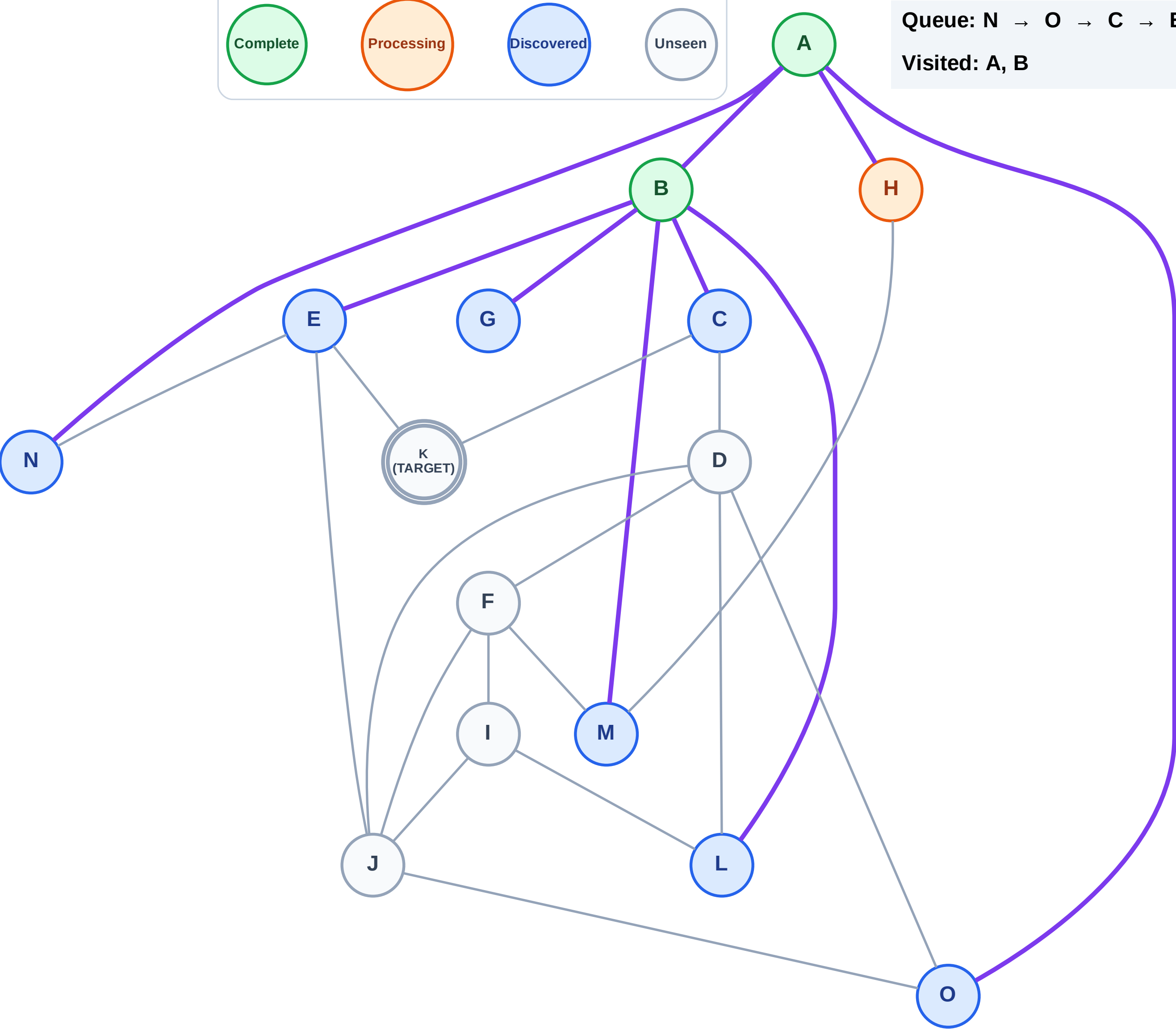
Processing

Discovered

Unseen

Queue: N → O → C → E → G → L → M

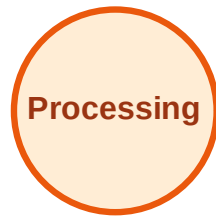
Visited: A, B



BFS Traversal

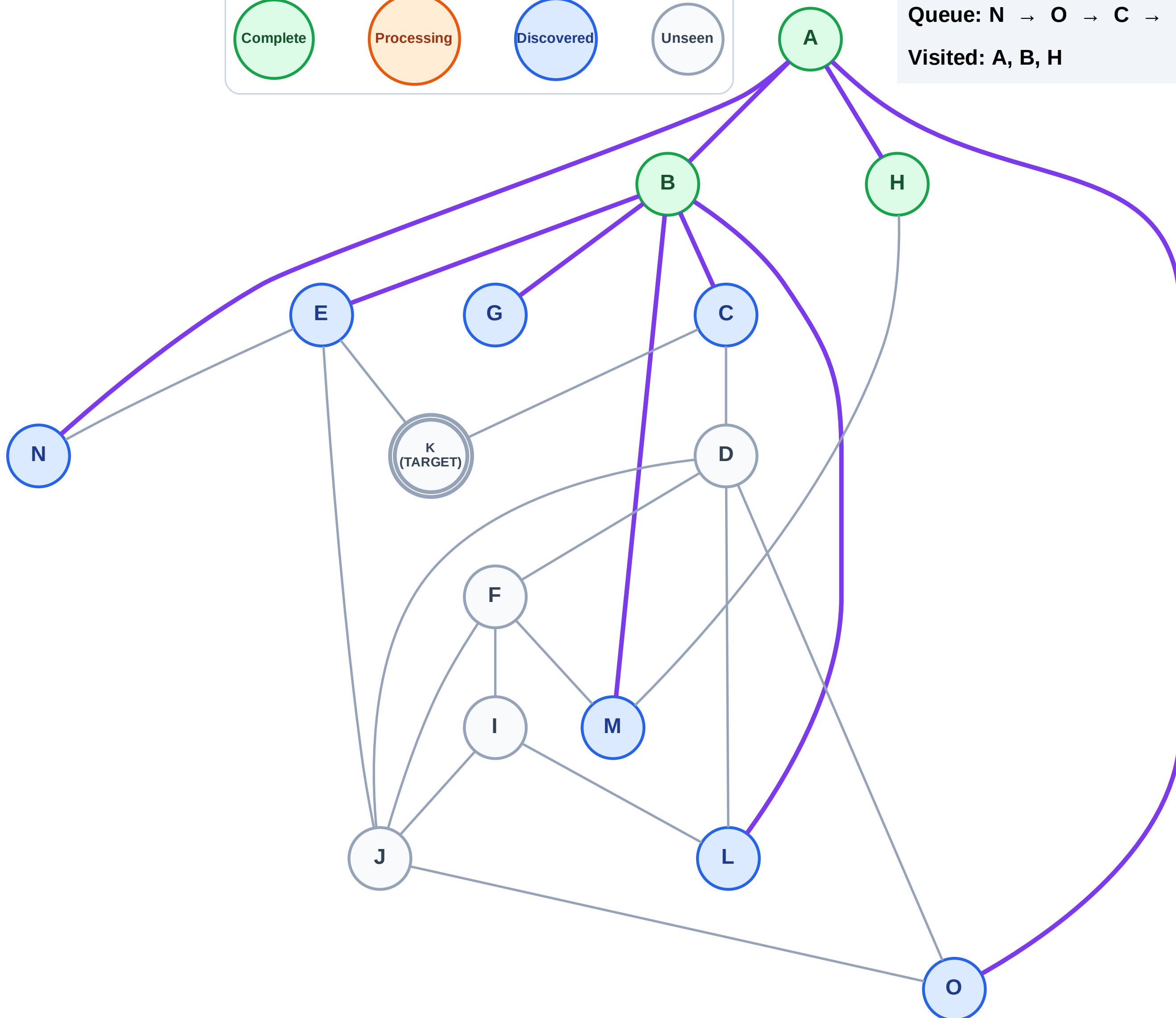
Step 7: No new neighbors from H

Legend



Queue: N → O → C → E → G → L → M

Visited: A, B, H



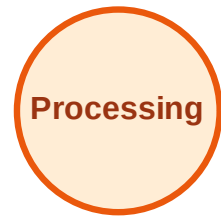
BFS Traversal

Step 8: Process node N

Legend



Complete



Processing



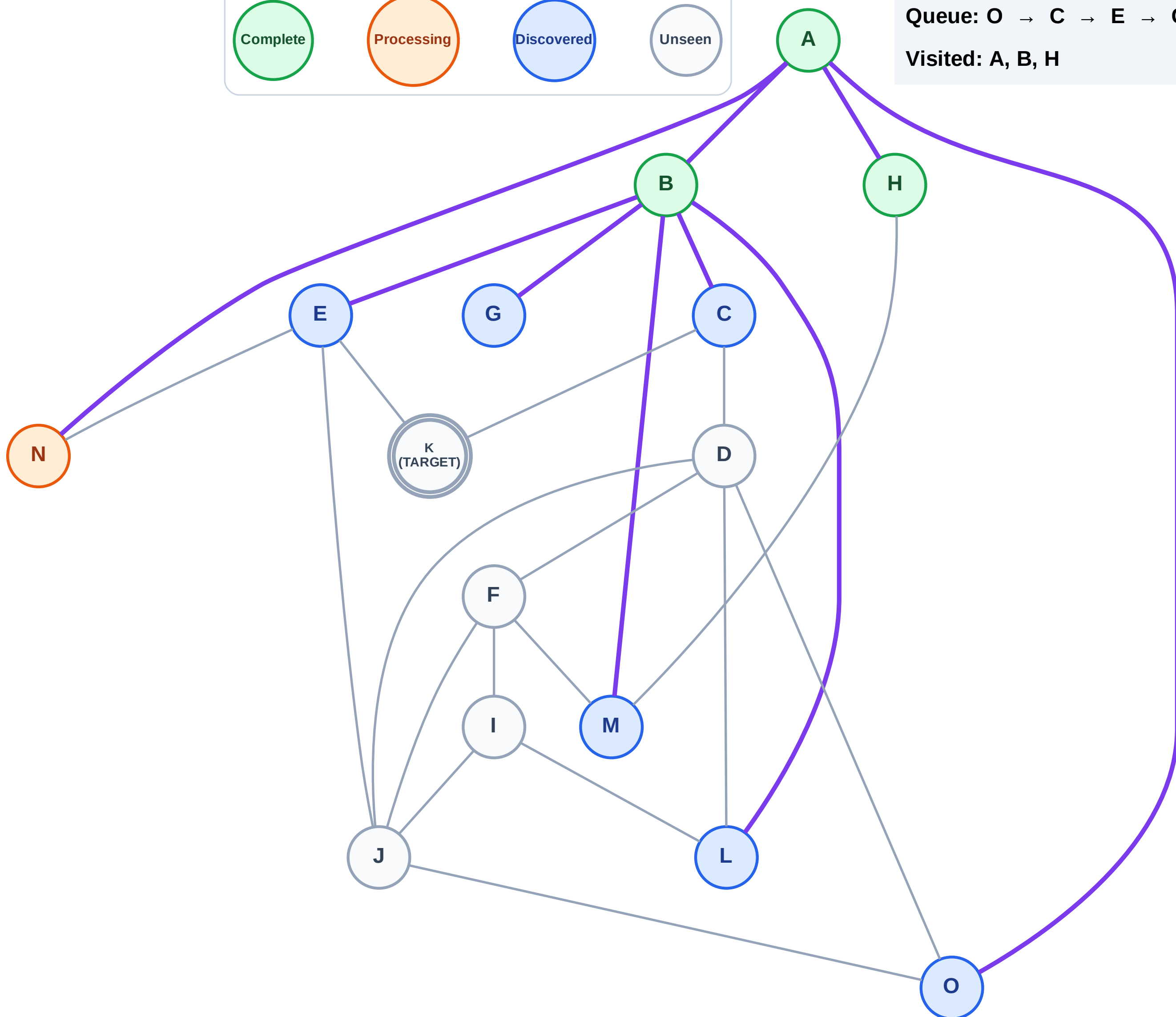
Discovered



Unseen

Queue: O → C → E → G → L → M

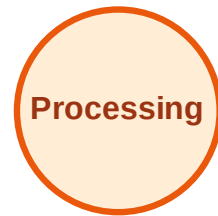
Visited: A, B, H



BFS Traversal

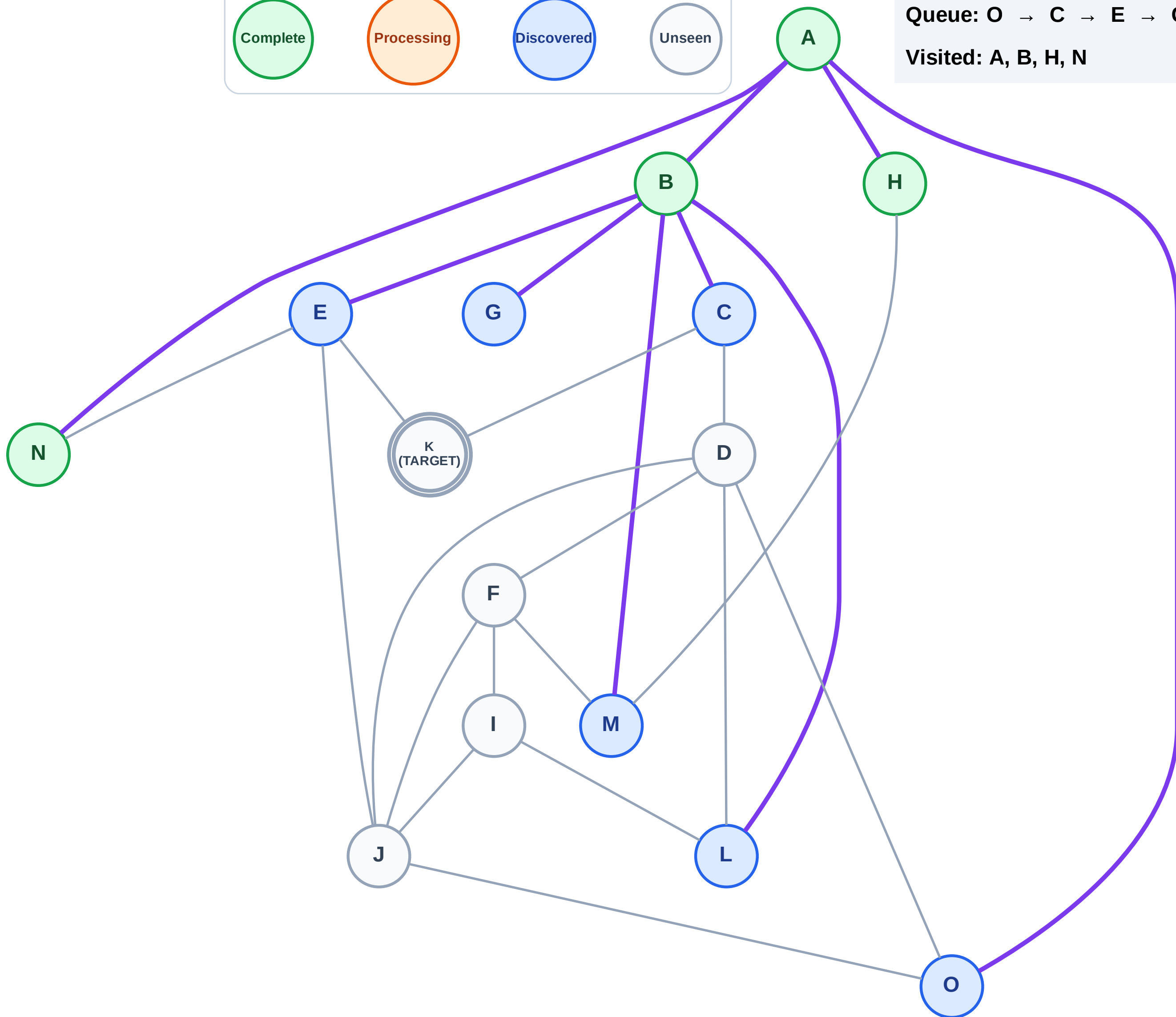
Step 9: No new neighbors from N

Legend



Queue: O → C → E → G → L → M

Visited: A, B, H, N



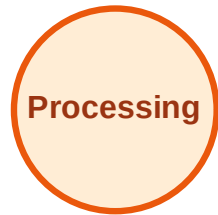
BFS Traversal

Step 10: Process node O

Legend



Complete



Processing



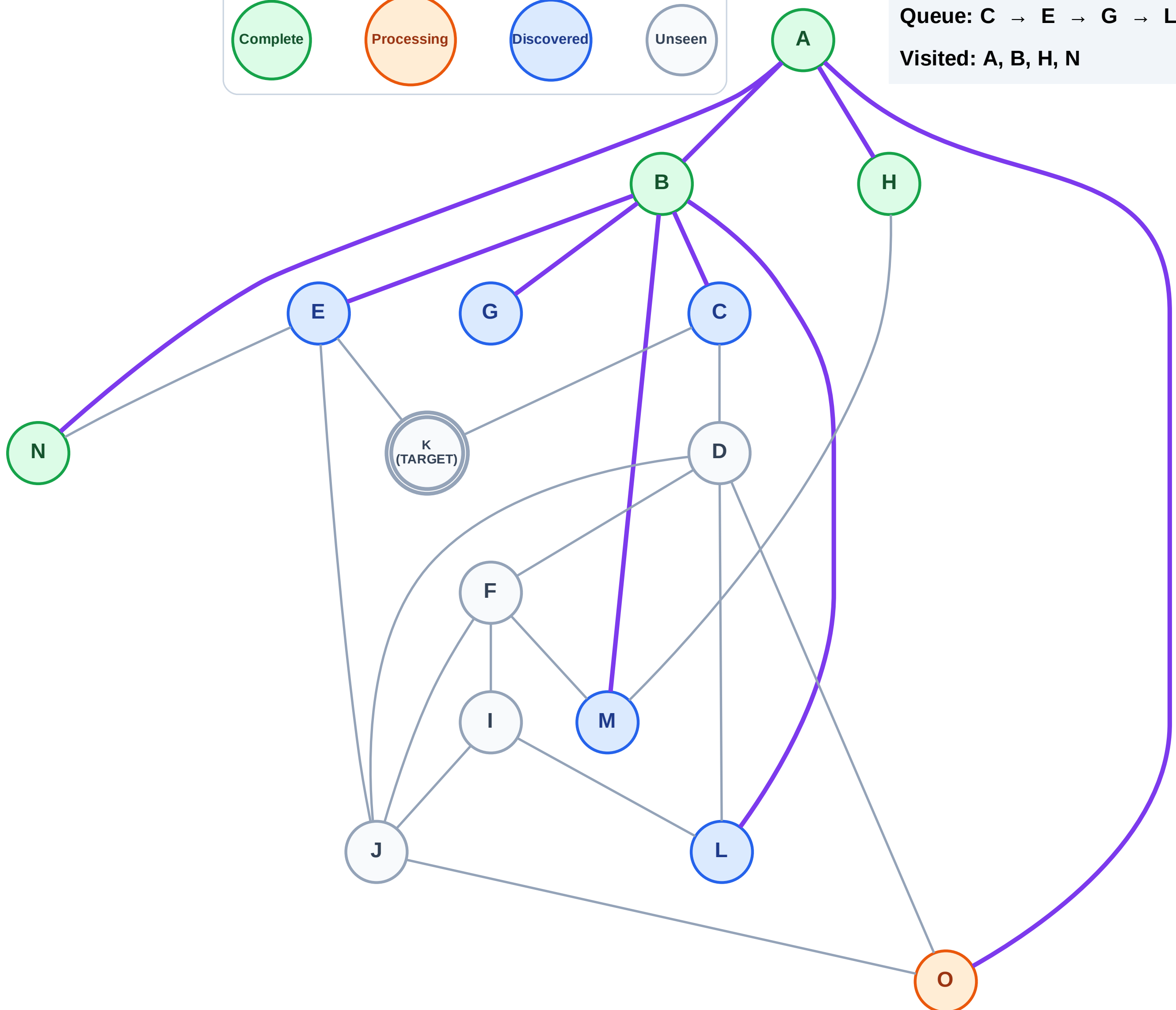
Discovered



Unseen

Queue: C → E → G → L → M

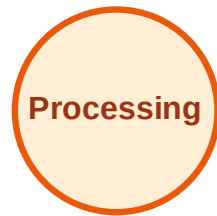
Visited: A, B, H, N



BFS Traversal

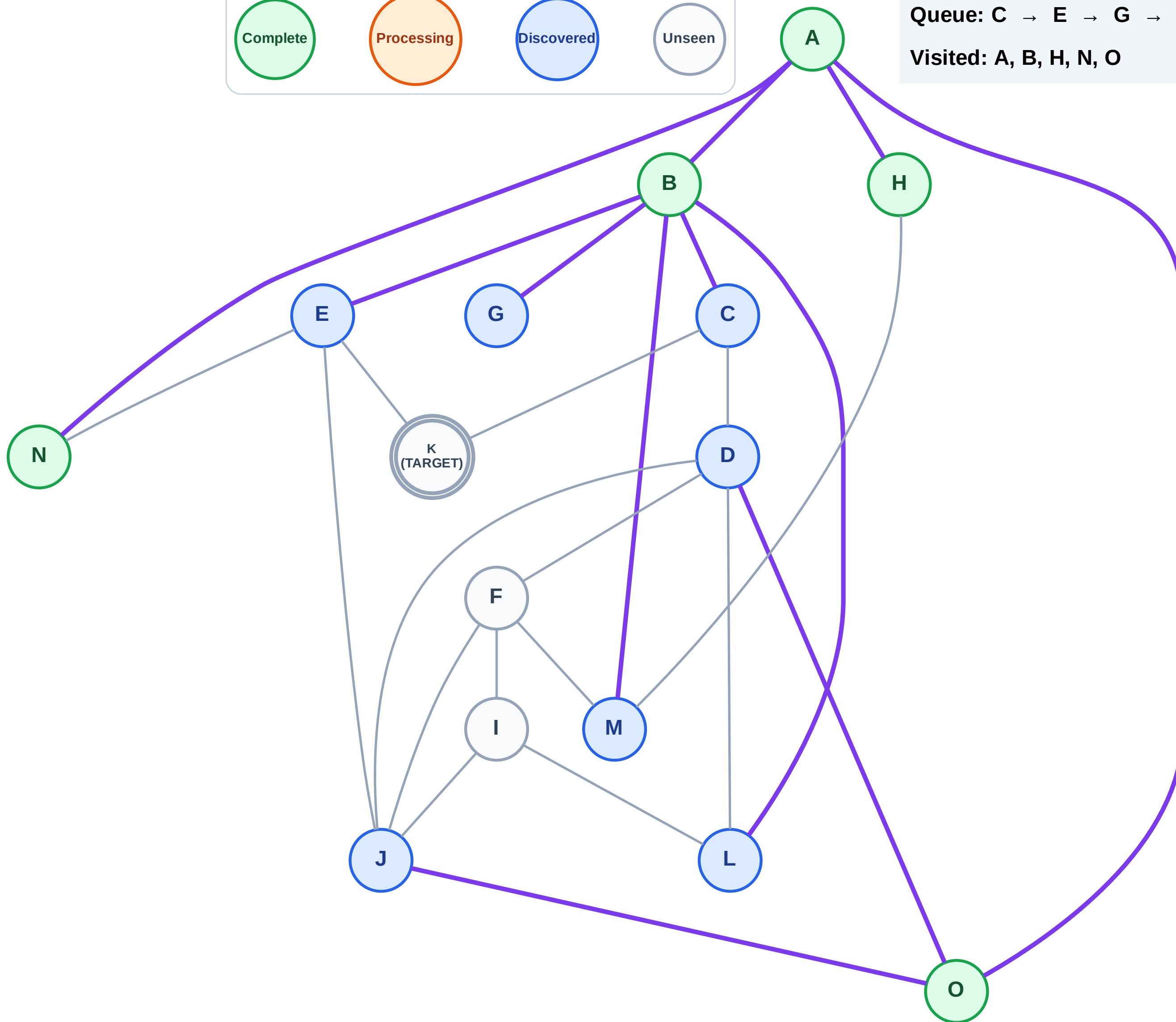
Step 11: Discover D, J from O

Legend



Queue: C → E → G → L → M → D → J

Visited: A, B, H, N, O



BFS Traversal

Step 12: Process node C

Legend



Complete



Processing



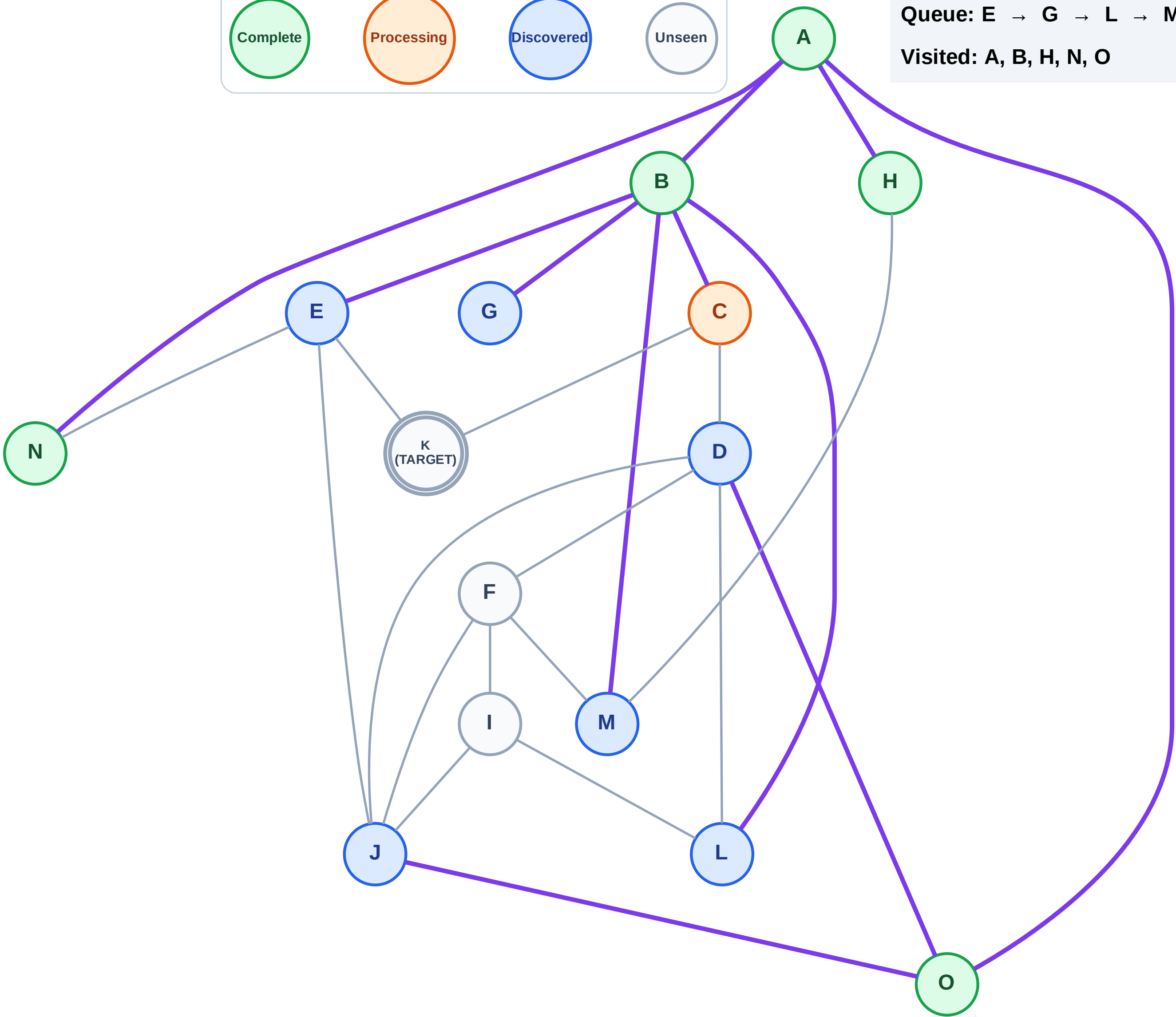
Discovered



Unseen

Queue: E → G → L → M → D → J

Visited: A, B, H, N, O



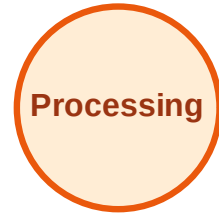
BFS Traversal

Step 13: Discover K from C

Legend



Complete



Processing



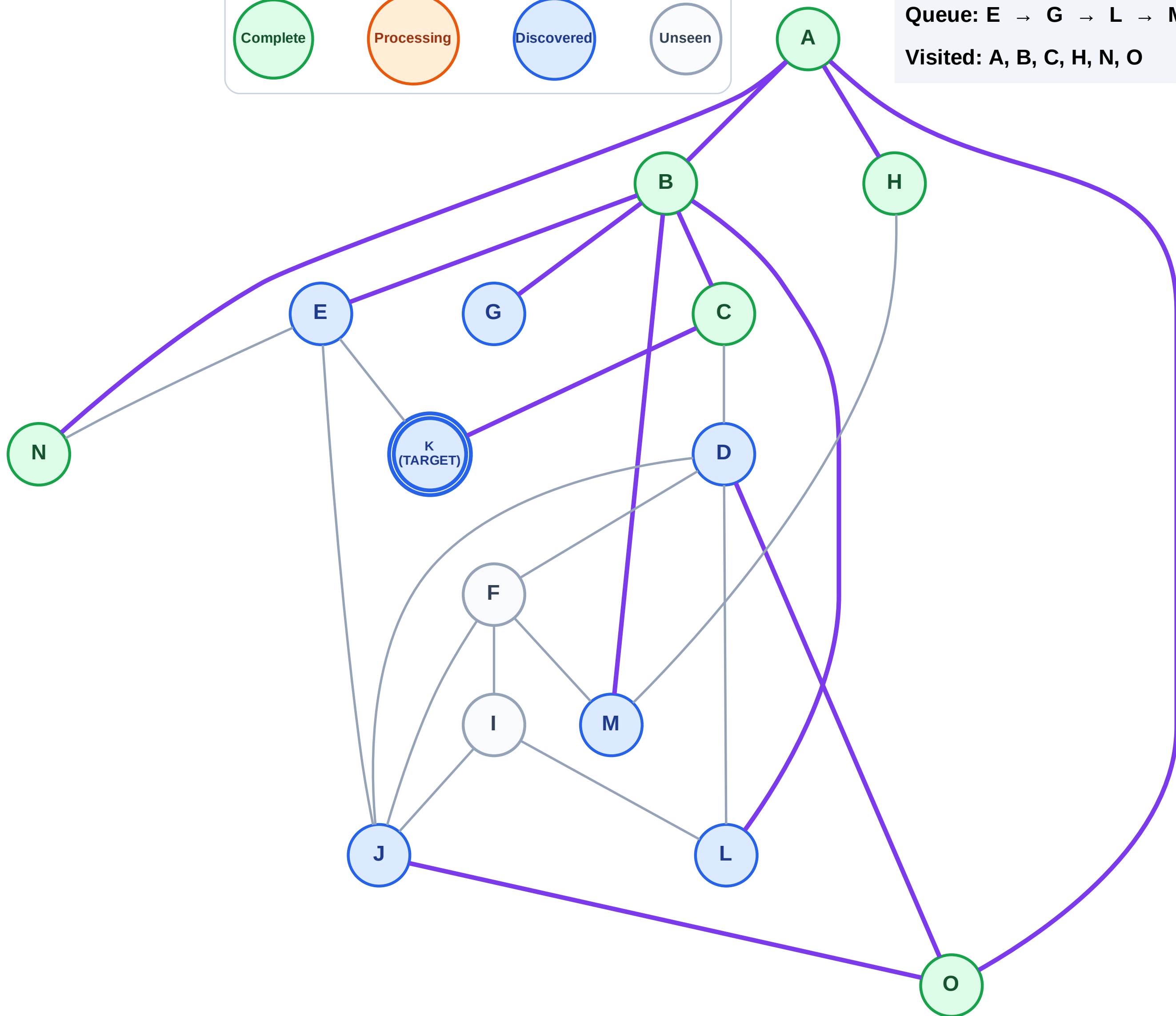
Discovered



Unseen

Queue: E → G → L → M → D → J → K

Visited: A, B, C, H, N, O



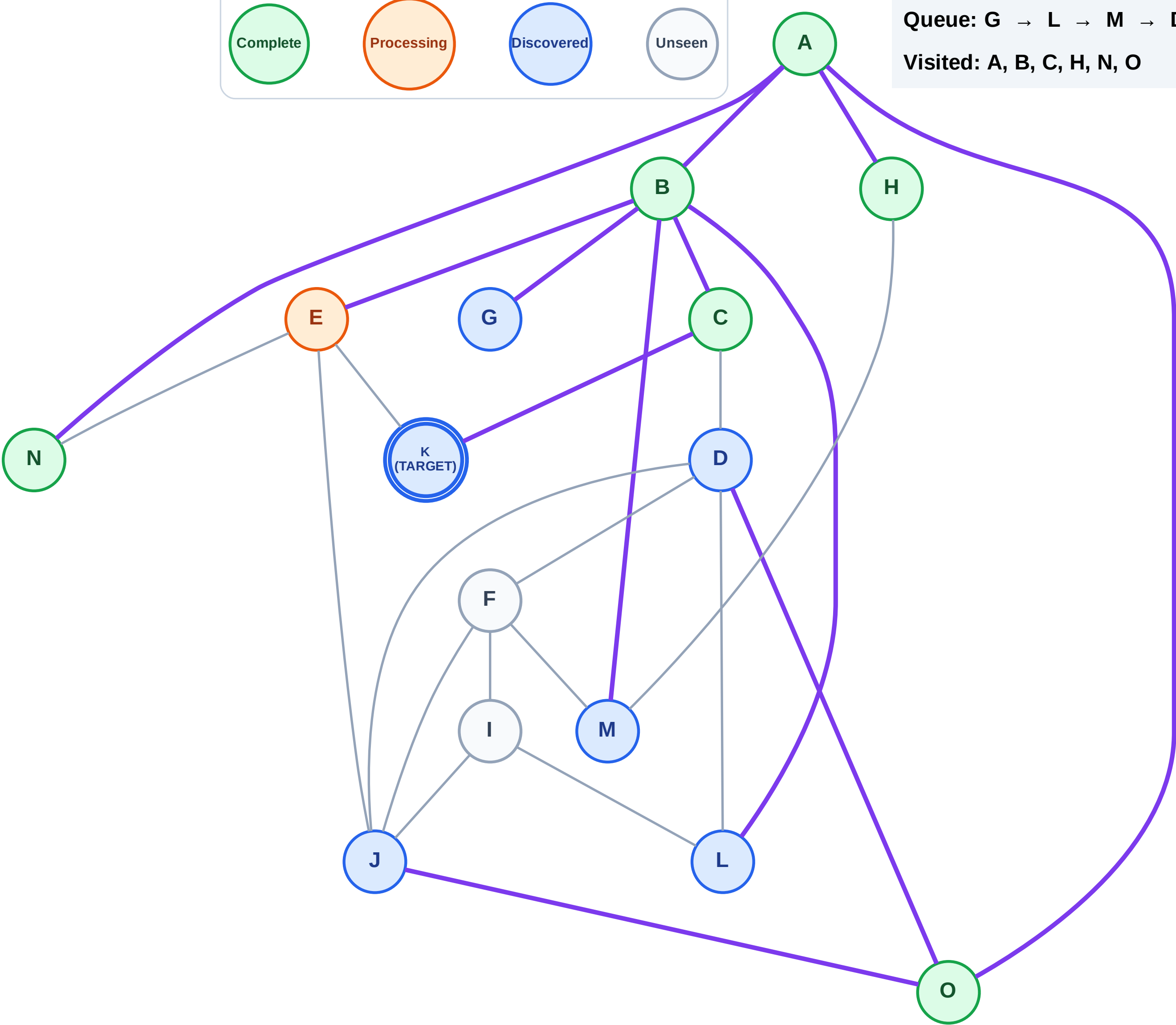
BFS Traversal

Step 14: Process node E

Legend



Queue: G → L → M → D → J → K
Visited: A, B, C, H, N, O



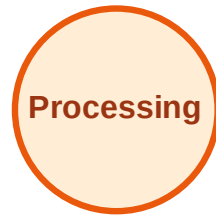
BFS Traversal

Step 15: No new neighbors from E

Legend



Complete



Processing



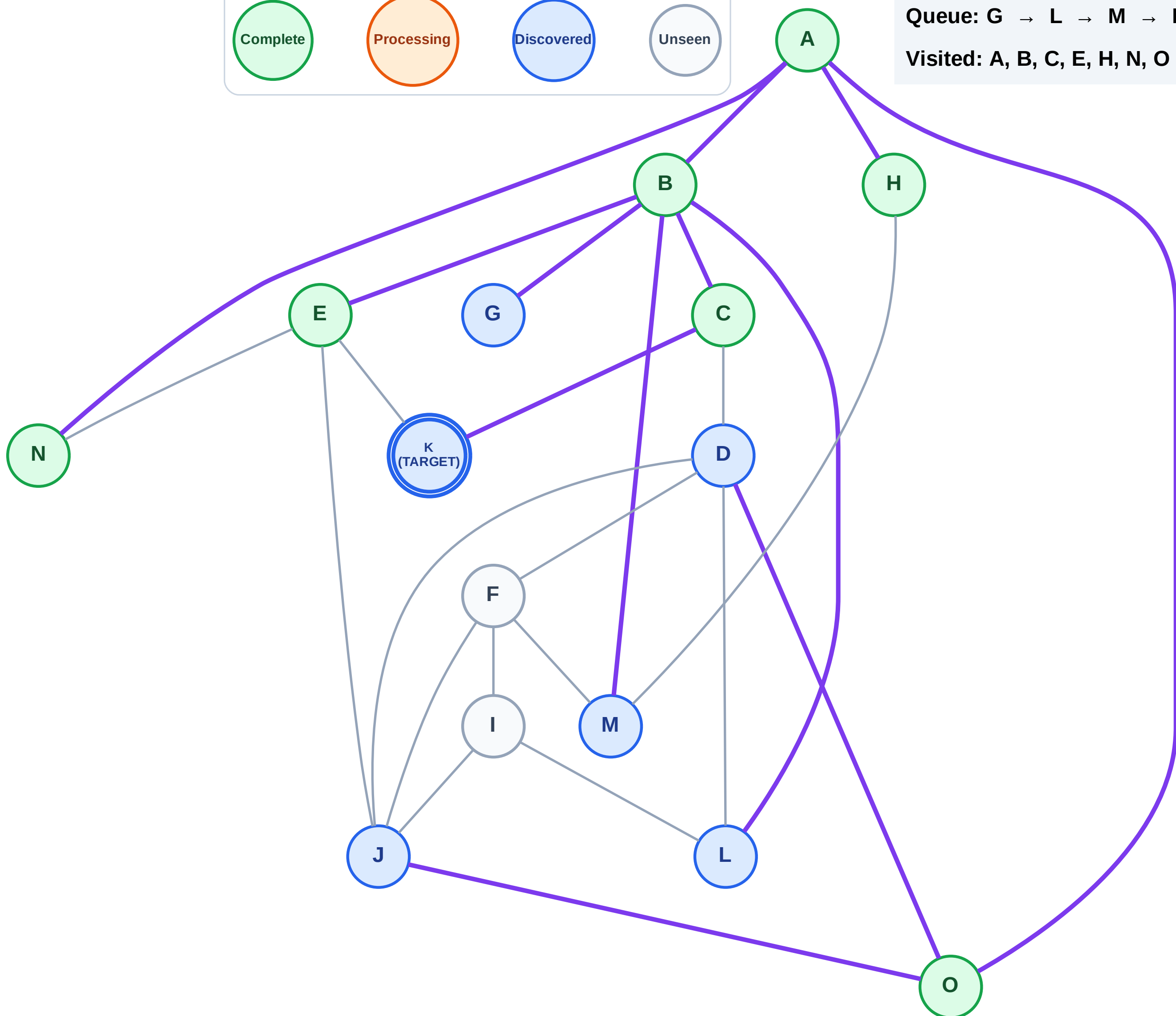
Discovered



Unseen

Queue: G → L → M → D → J → K

Visited: A, B, C, E, H, N, O



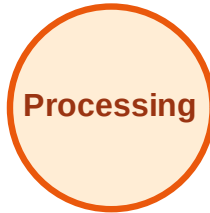
BFS Traversal

Step 16: Process node G

Legend



Complete



Processing



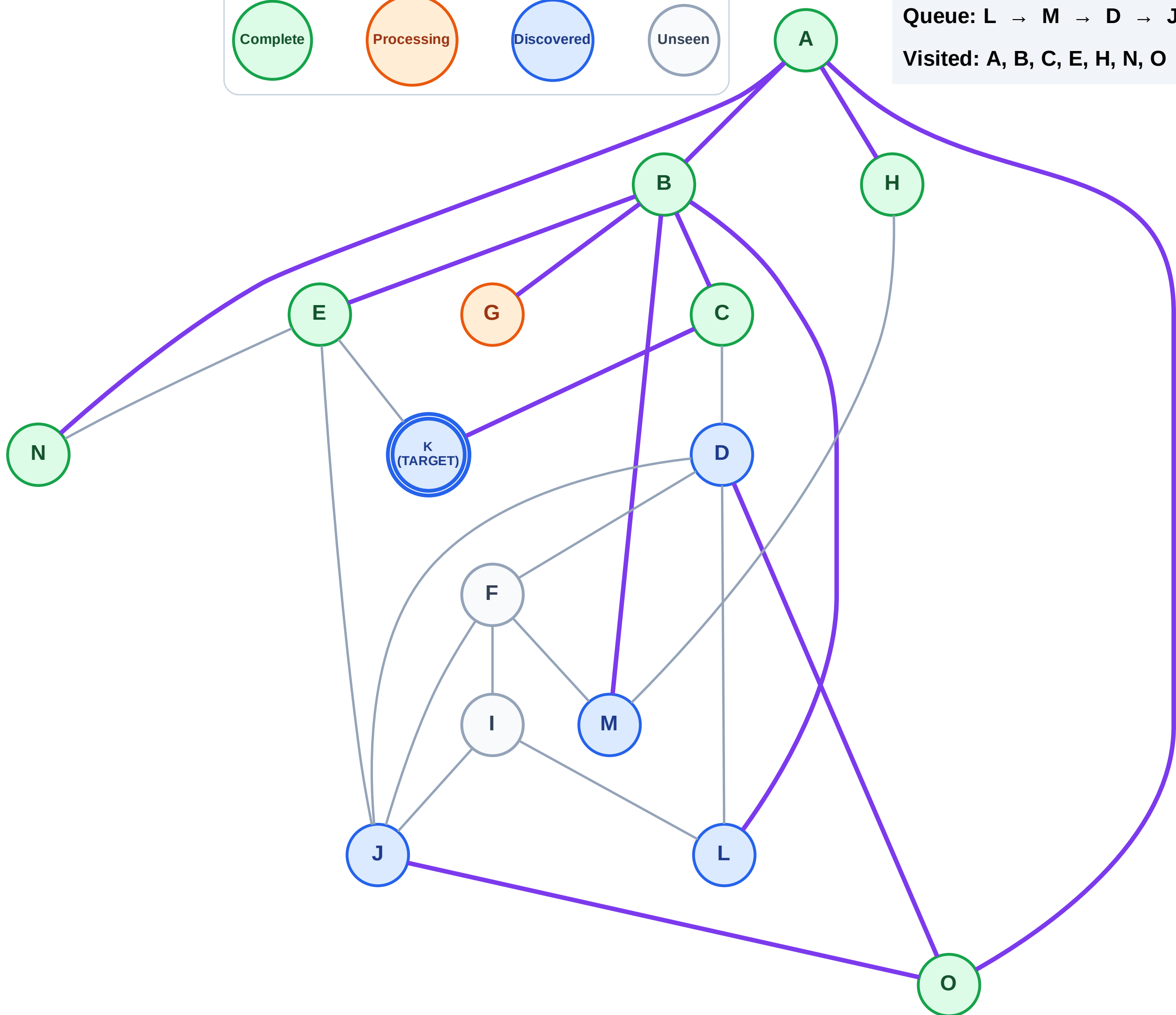
Discovered



Unseen

Queue: L → M → D → J → K

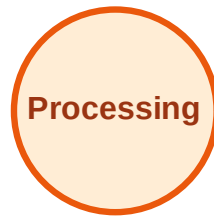
Visited: A, B, C, E, H, N, O



BFS Traversal

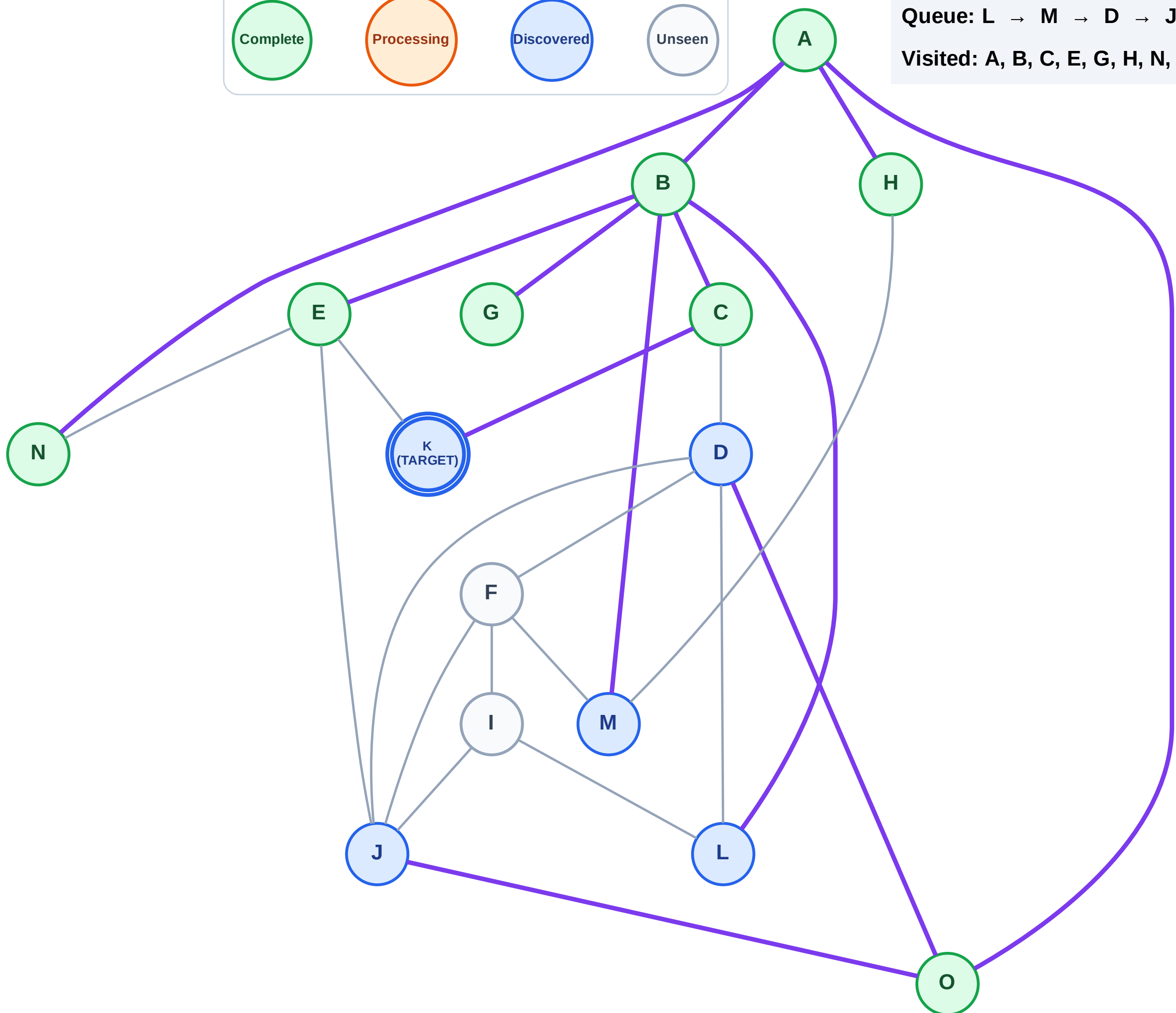
Step 17: No new neighbors from G

Legend



Queue: L → M → D → J → K

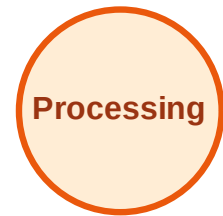
Visited: A, B, C, E, G, H, N, O



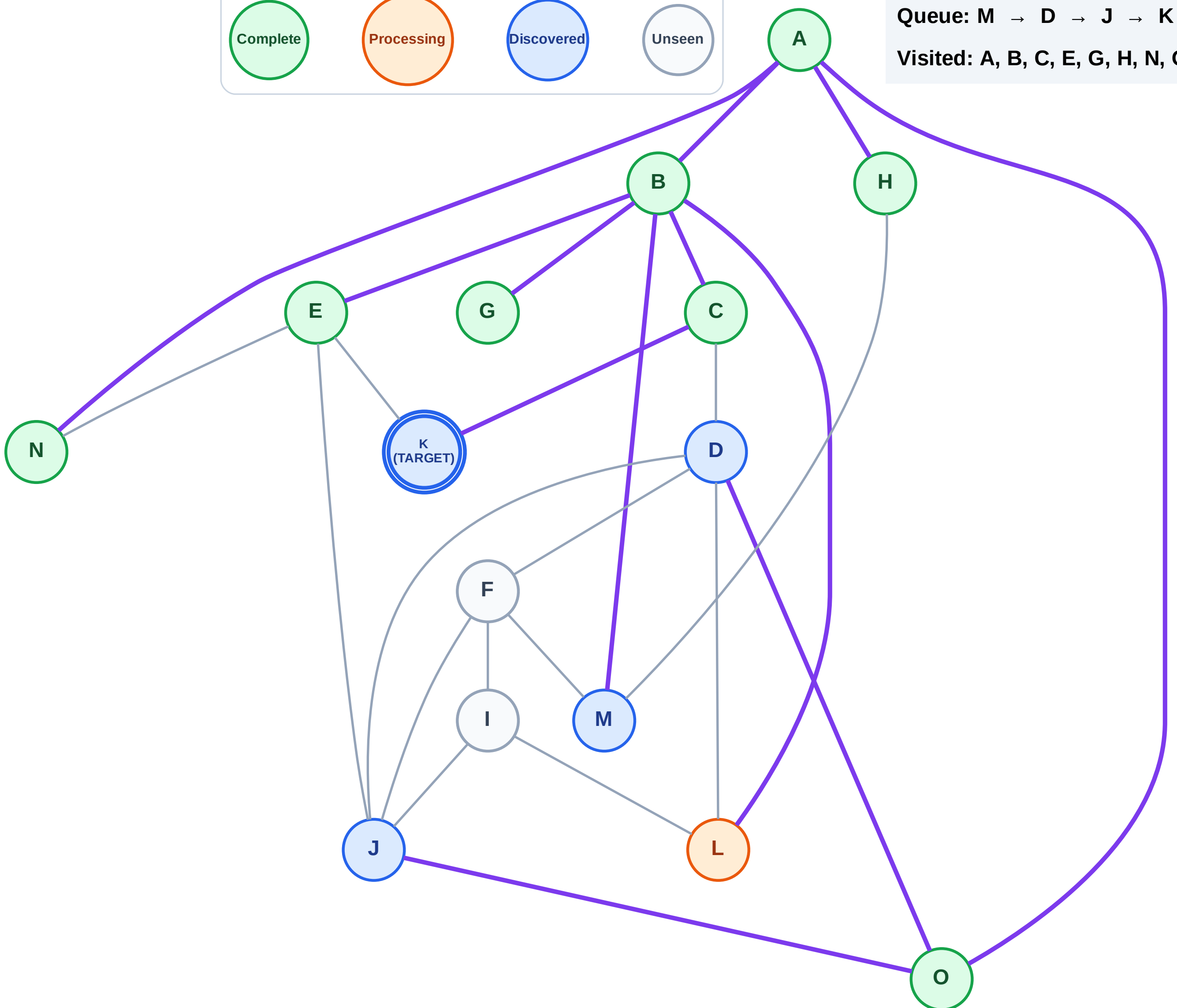
BFS Traversal

Step 18: Process node L

Legend



Queue: M → D → J → K
Visited: A, B, C, E, G, H, N, O



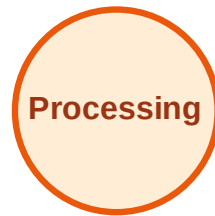
BFS Traversal

Step 19: Discover I from L

Legend



Complete



Processing



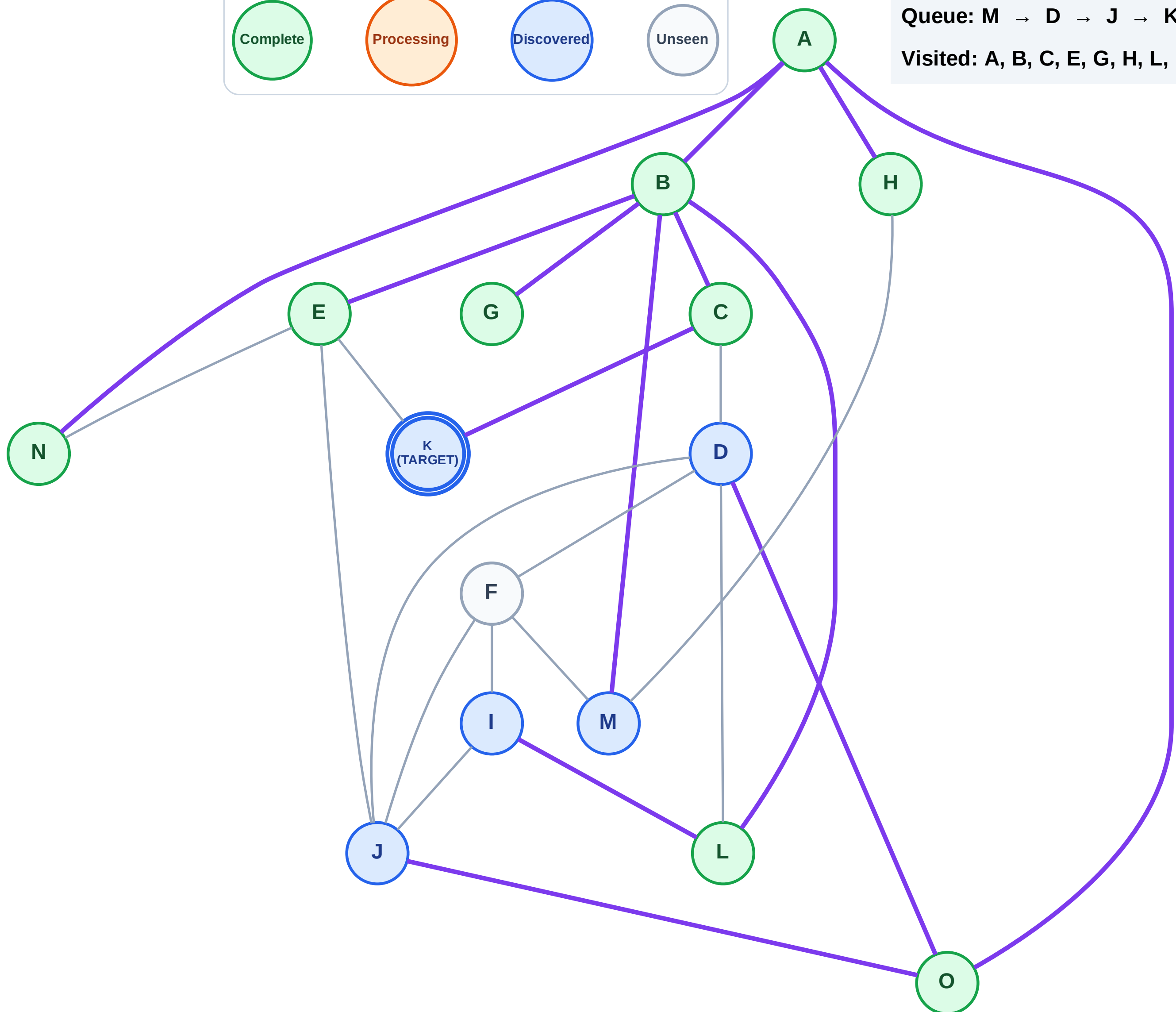
Discovered



Unseen

Queue: M → D → J → K → I

Visited: A, B, C, E, G, H, L, N, O



BFS Traversal

Step 20: Process node M

Legend



Complete



Processing



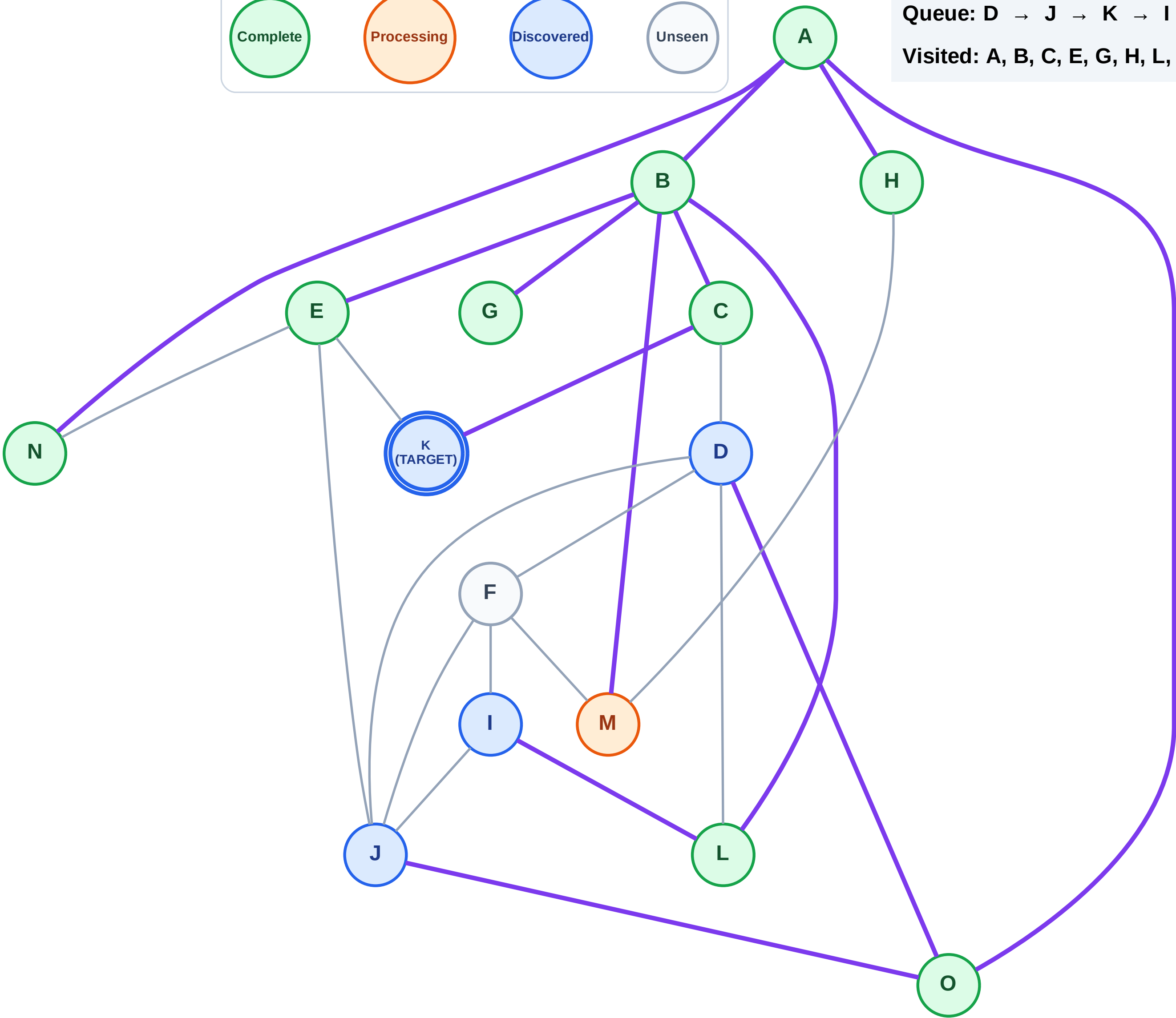
Discovered



Unseen

Queue: D → J → K → I

Visited: A, B, C, E, G, H, L, N, O



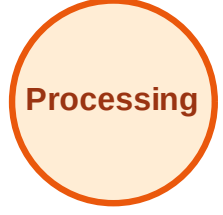
BFS Traversal

Step 21: Discover F from M

Legend



Complete



Processing

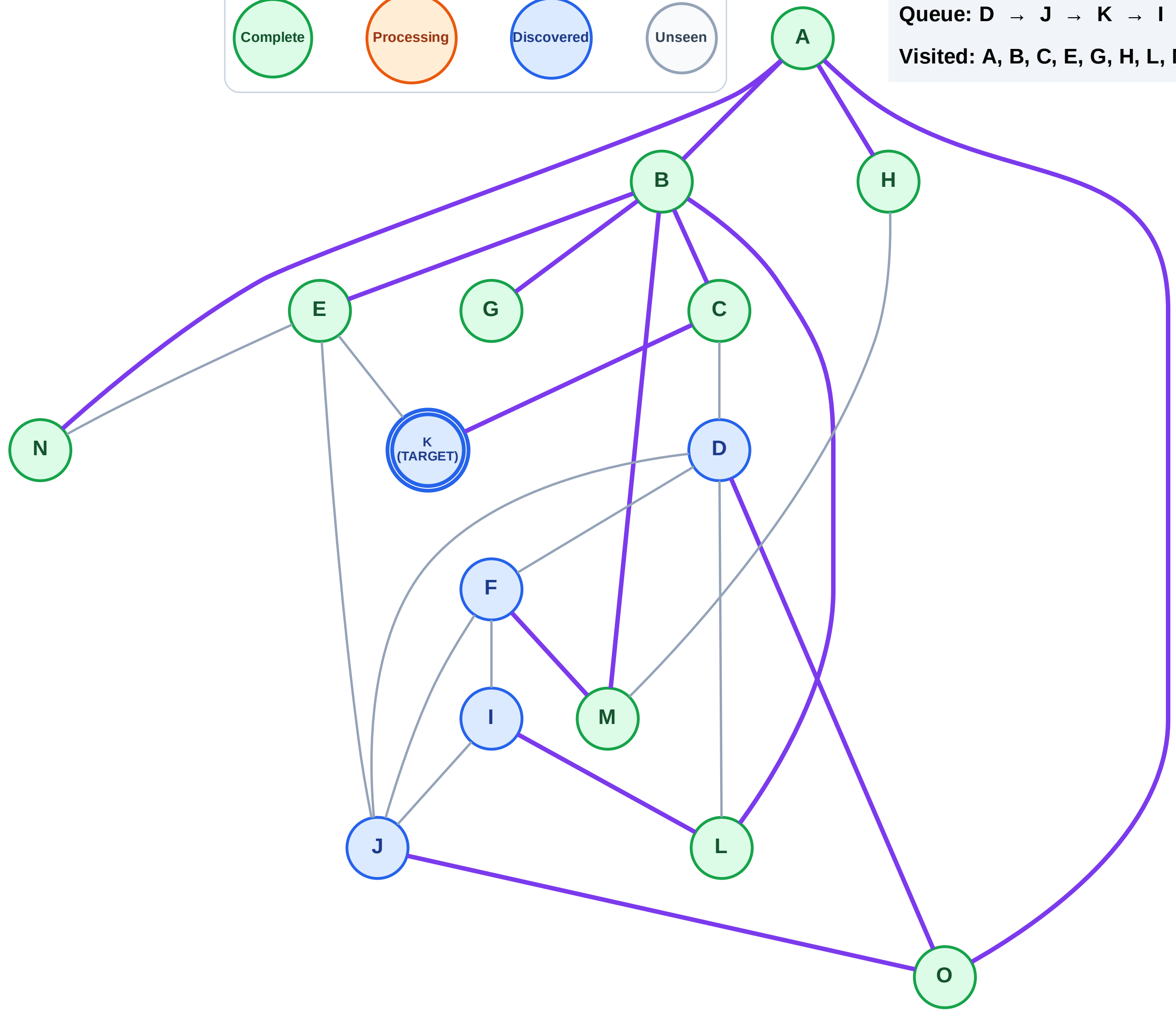


Discovered



Unseen

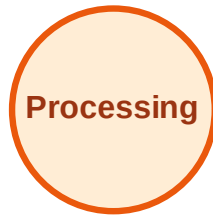
Queue: D → J → K → I → F
Visited: A, B, C, E, G, H, L, M, N, O



BFS Traversal

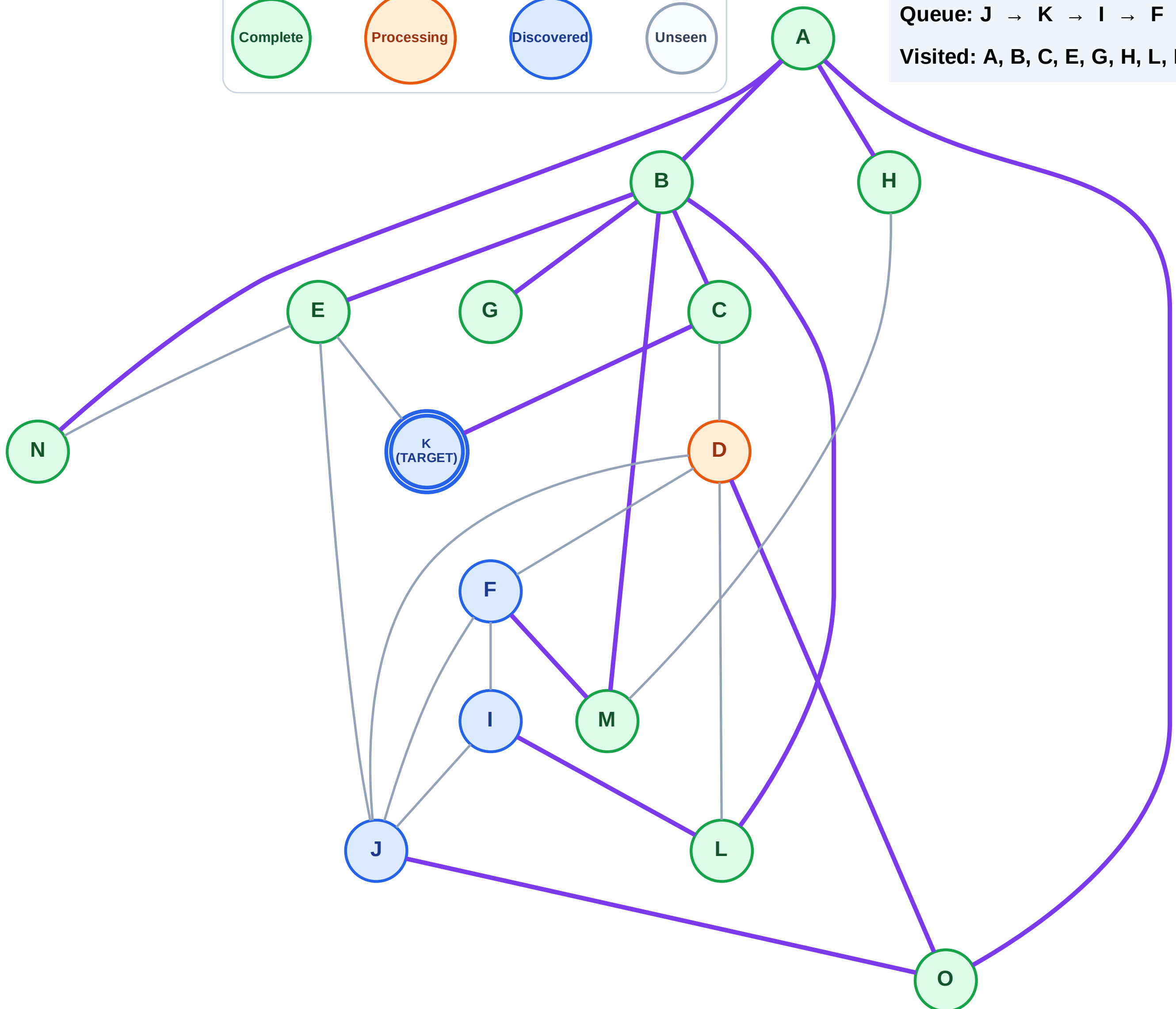
Step 22: Process node D

Legend



Queue: J → K → I → F

Visited: A, B, C, E, G, H, L, M, N, O



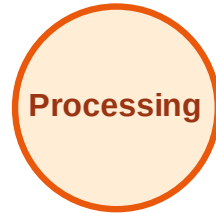
BFS Traversal

Step 23: No new neighbors from D

Legend



Complete



Processing



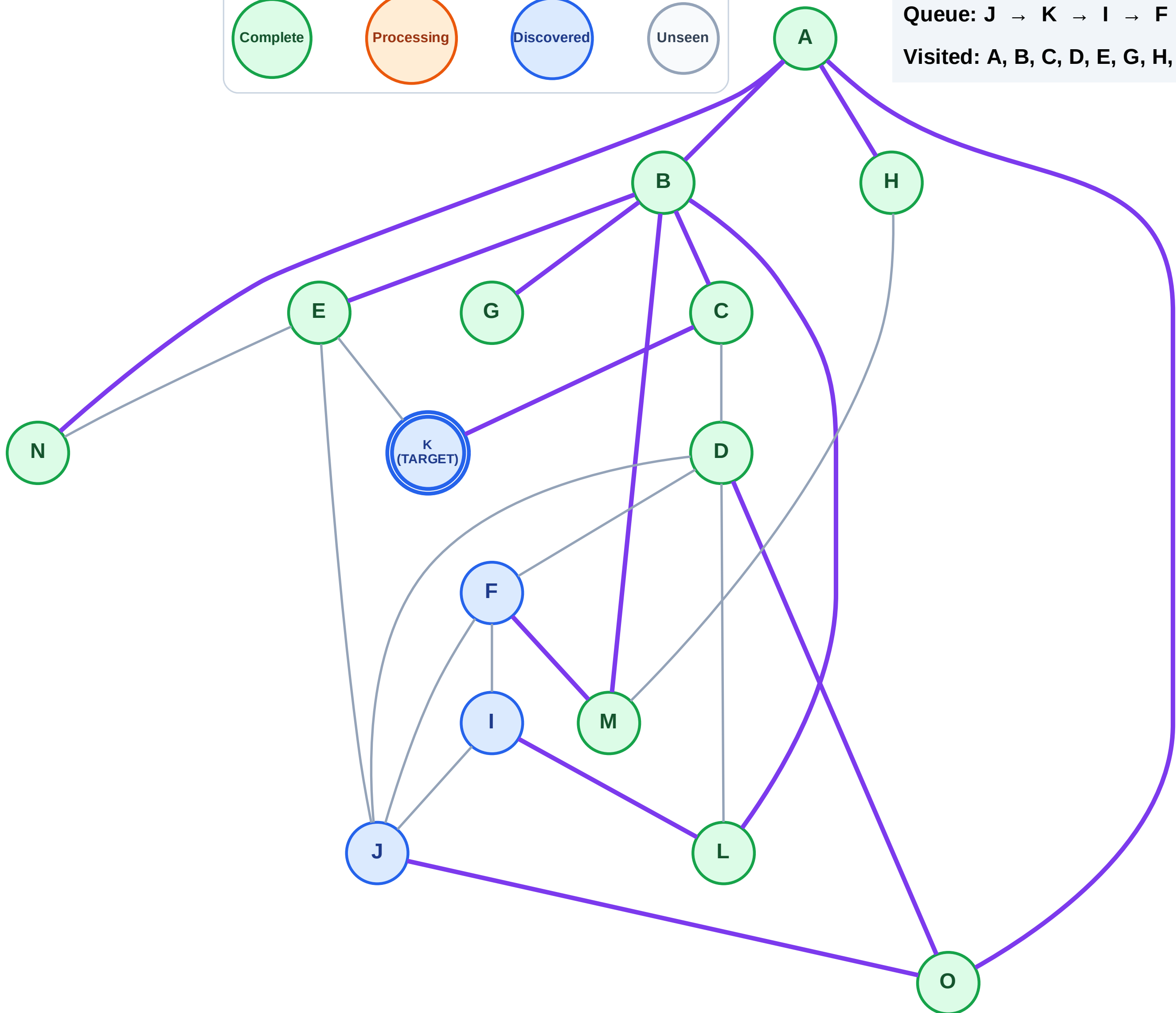
Discovered



Unseen

Queue: J → K → I → F

Visited: A, B, C, D, E, G, H, L, M, N, O



BFS Traversal

Step 24: Process node J

Legend

Complete

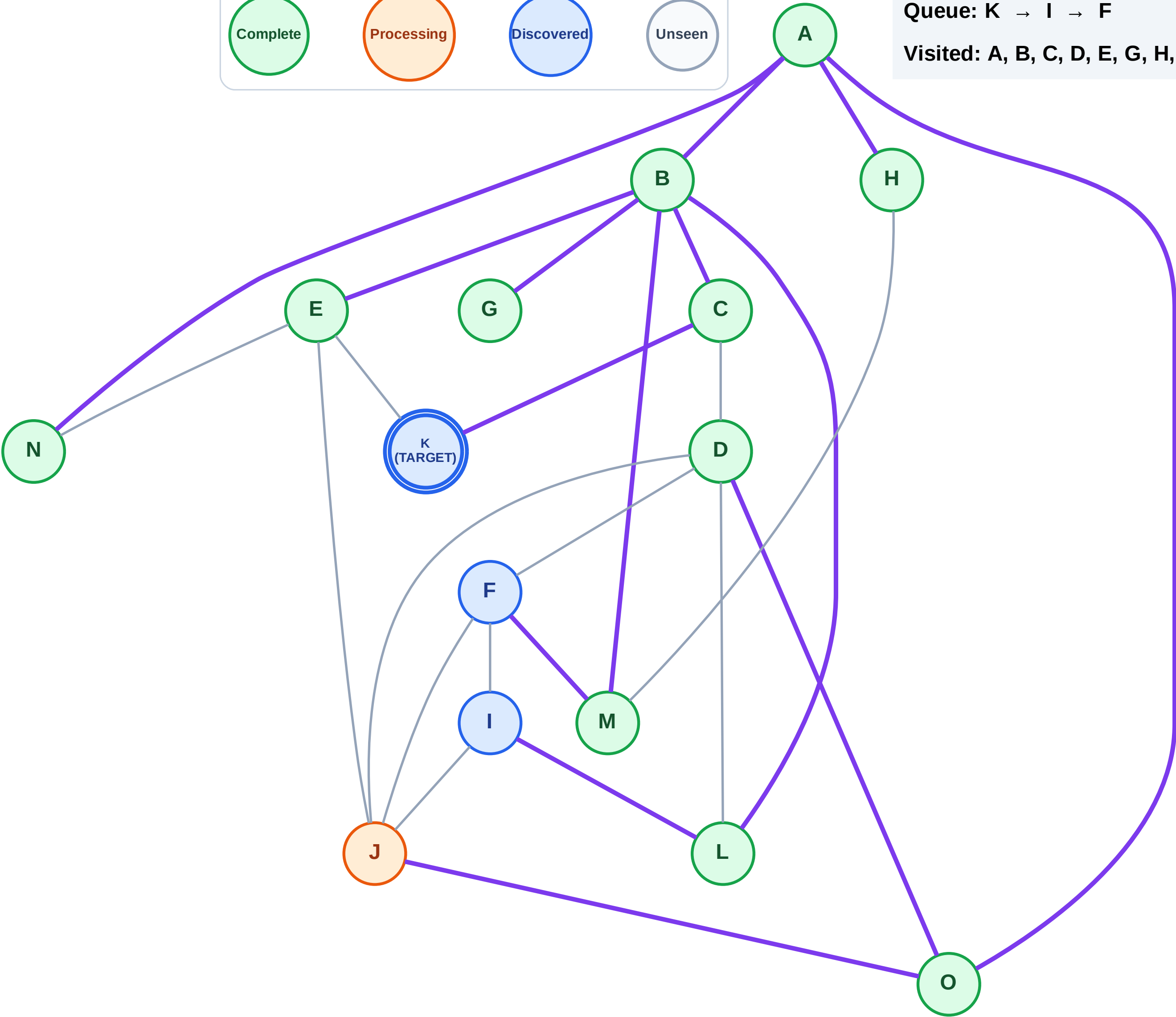
Processing

Discovered

Unseen

Queue: K → I → F

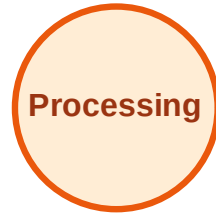
Visited: A, B, C, D, E, G, H, L, M, N, O



BFS Traversal

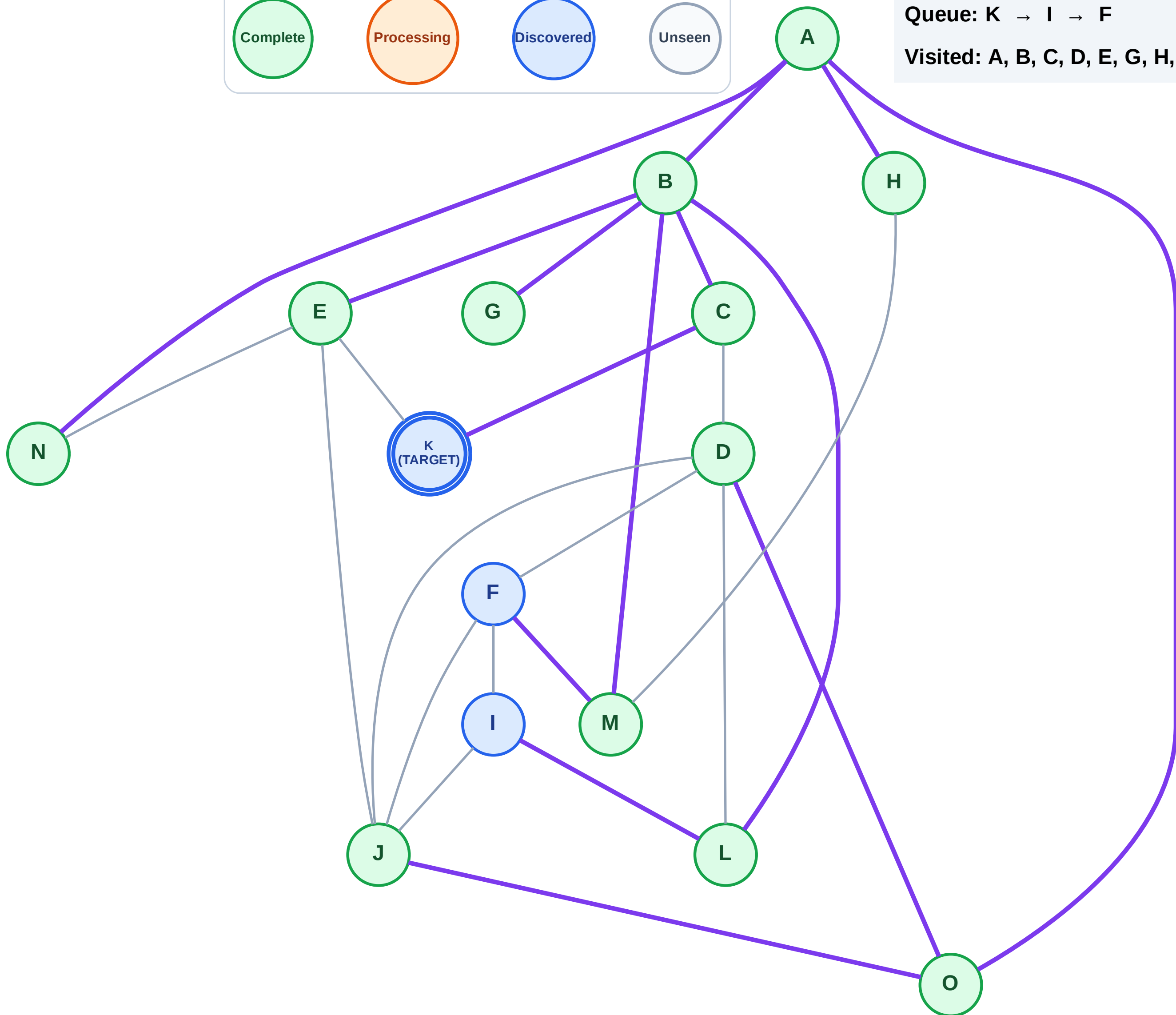
Step 25: No new neighbors from J

Legend



Queue: K → I → F

Visited: A, B, C, D, E, G, H, J, L, M, N, O



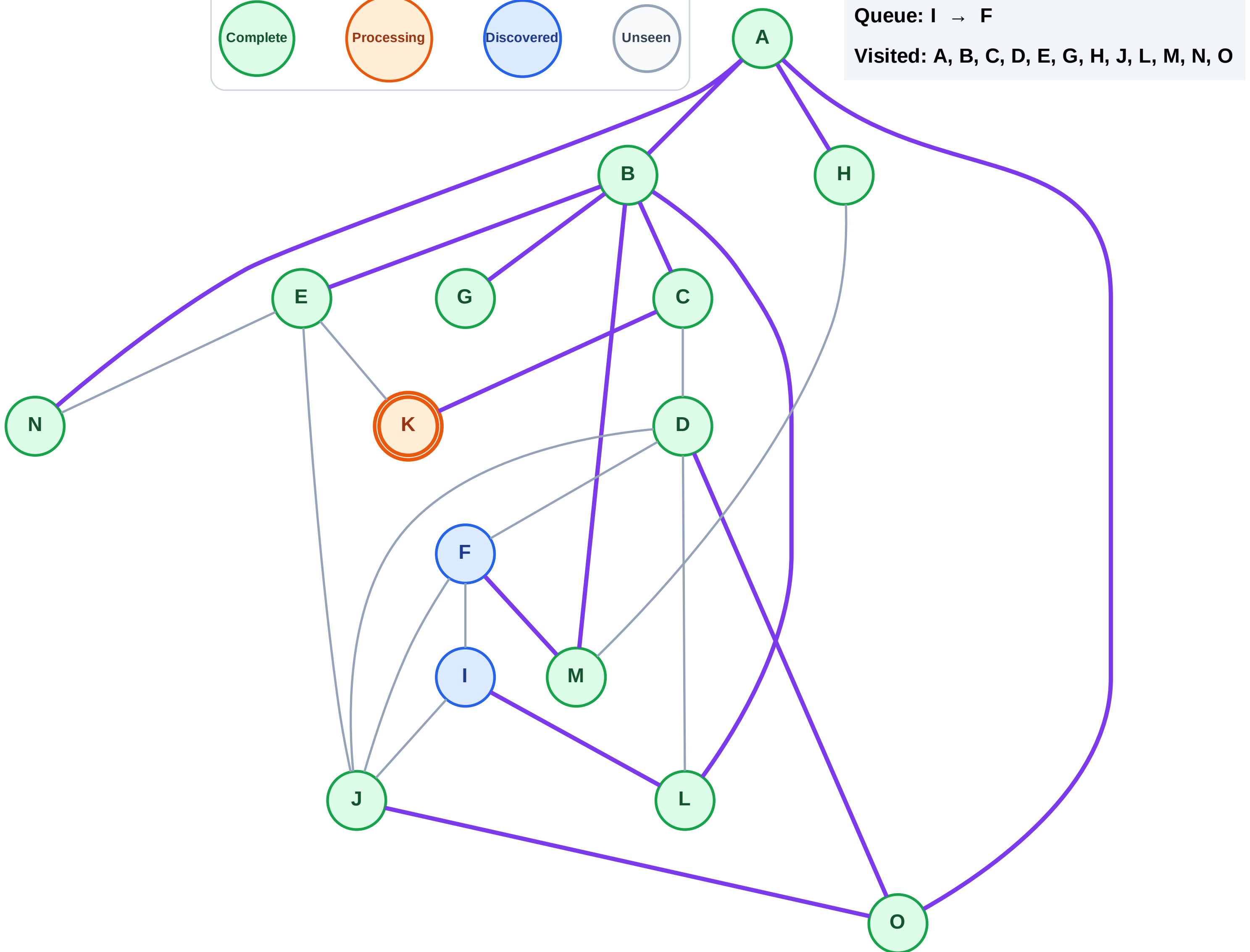
Step 26: Process node K

Page 10 of 10

Complete Processing Discovered Unseen

Unseen

Visited: A, B, C, D, E, G, H, J, L, M, N, O



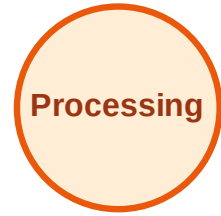
BFS Traversal

Step 27: TARGET NODE K REACHED! Path (3 hops): A → B → C → K

Legend



Complete



Processing



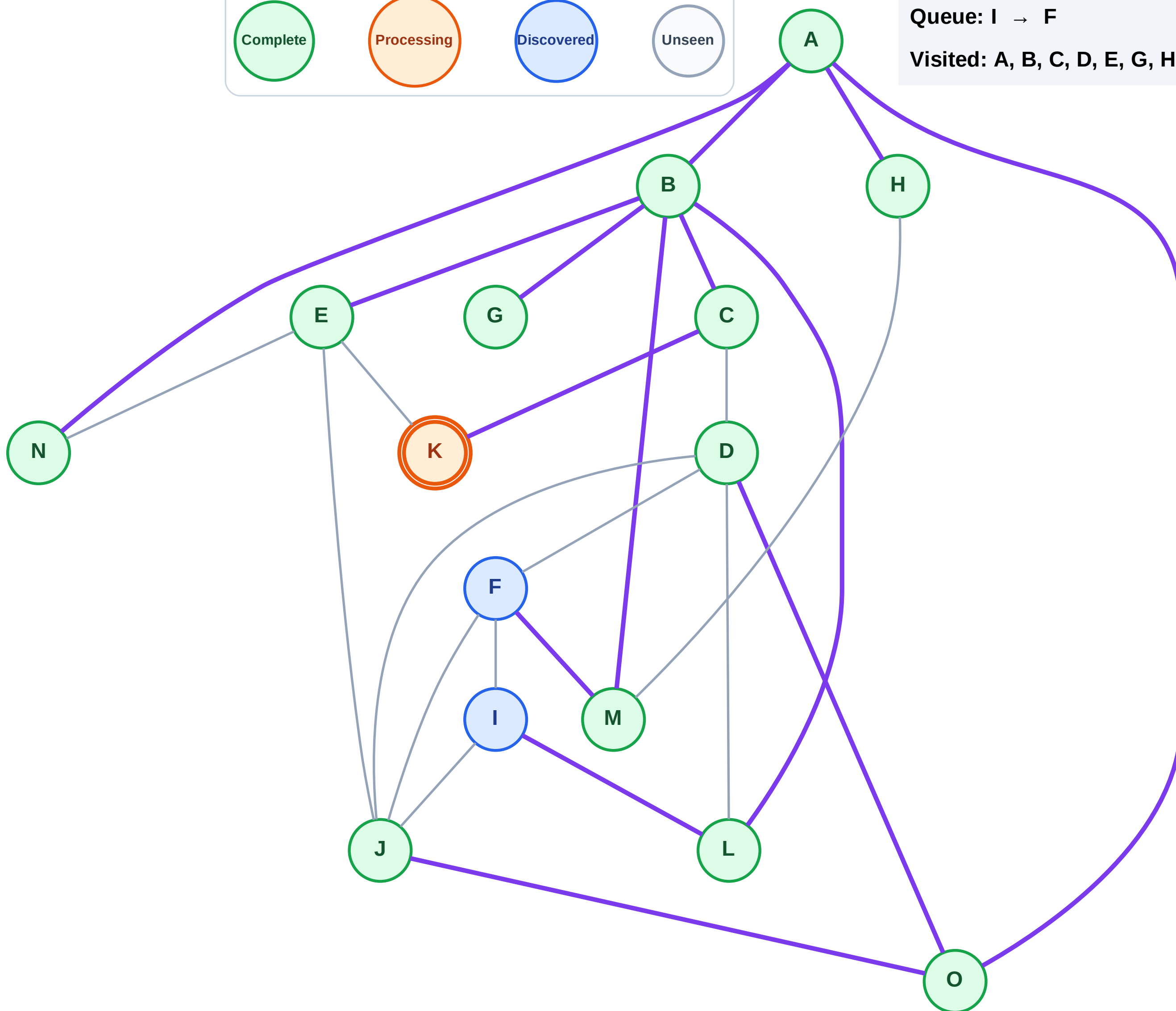
Discovered



Unseen

Queue: I → F

Visited: A, B, C, D, E, G, H, J, K, L, M, N, O



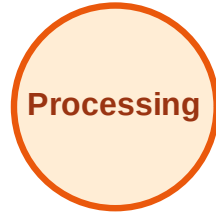
DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing



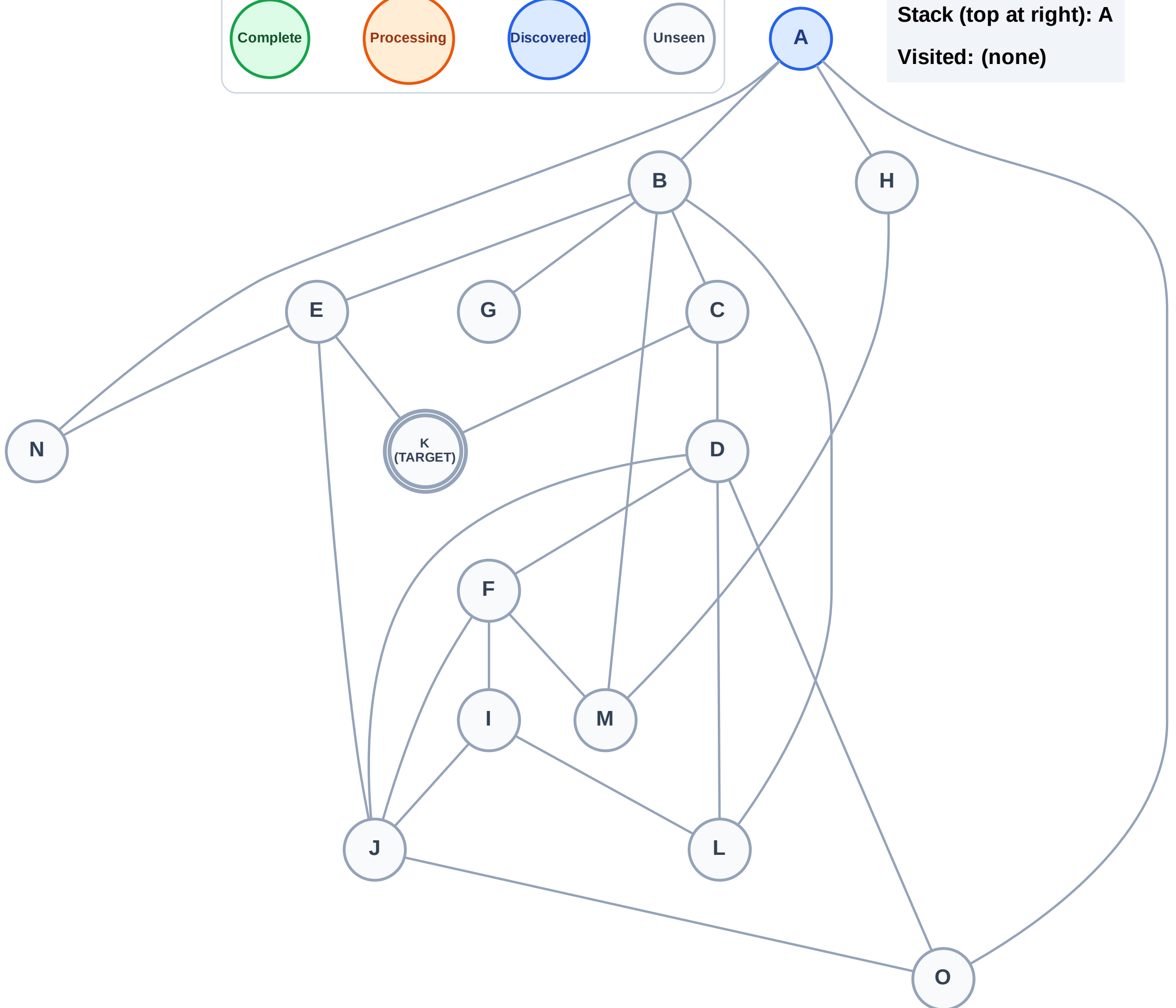
Discovered



Unseen

Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

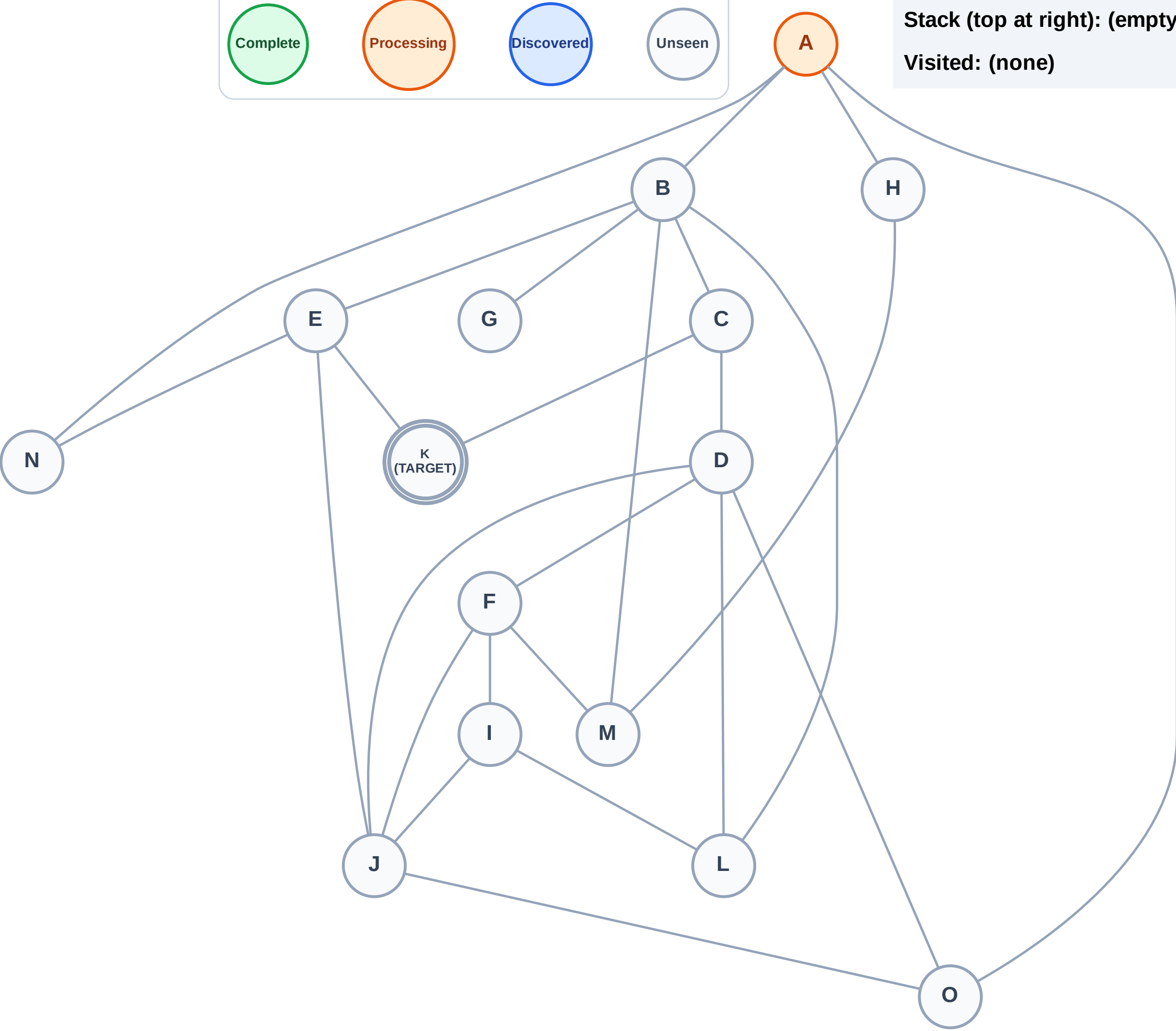
Complete

Processing

Discovered

Unseen

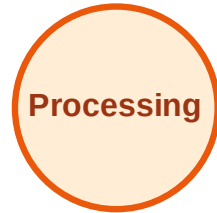
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

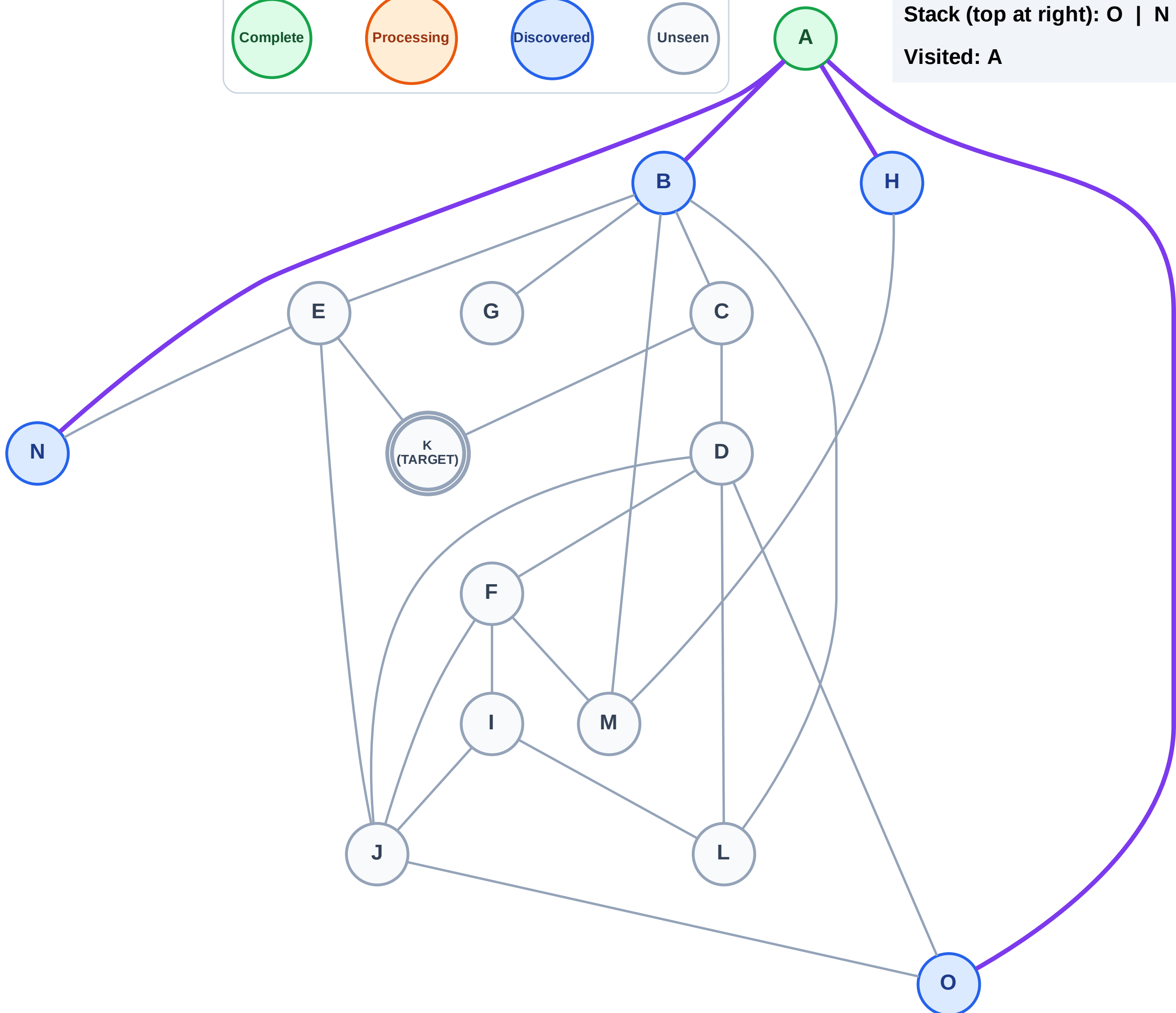
Step 3: Discover O, N, H, B from A

Legend



Stack (top at right): O | N | H | B

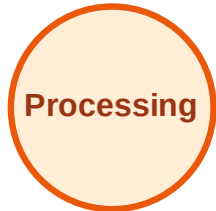
Visited: A



DFS Traversal

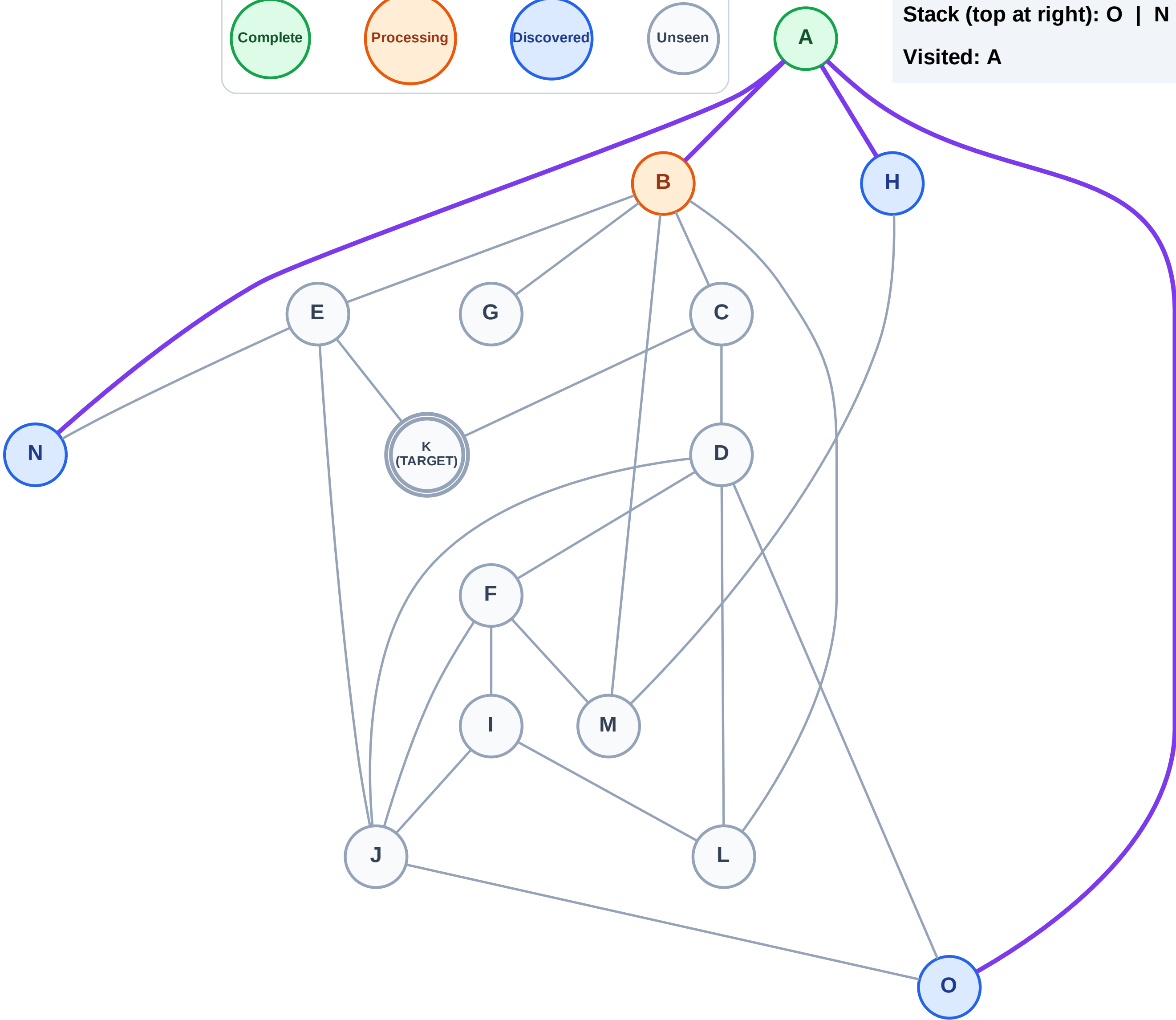
Step 4: Process node B

Legend



Stack (top at right): O | N | H

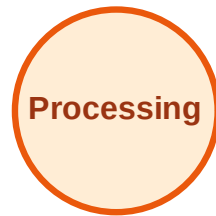
Visited: A



DFS Traversal

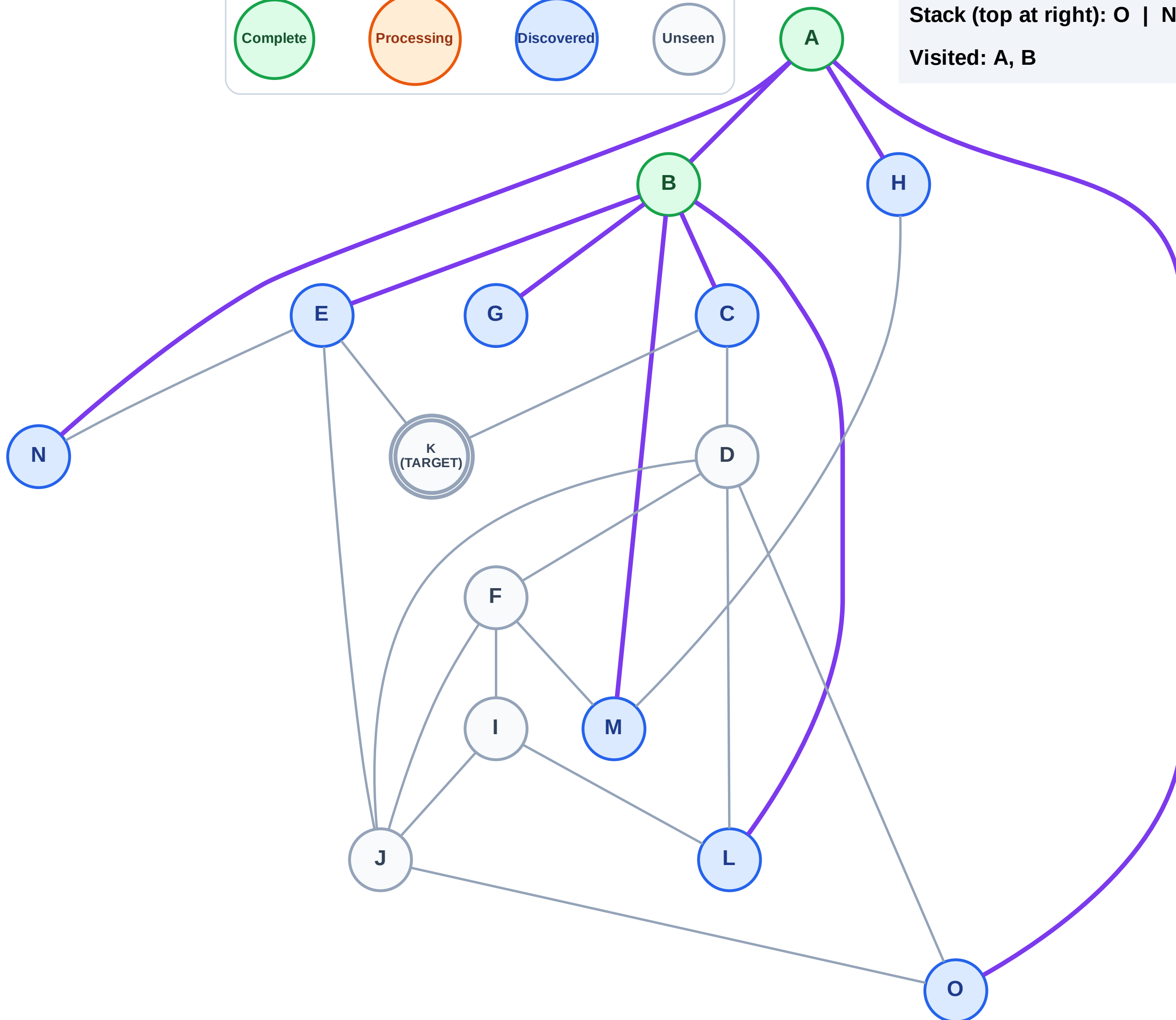
Step 5: Discover M, L, G, E, C from B

Legend



Stack (top at right): O | N | H | M | L | G | E | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

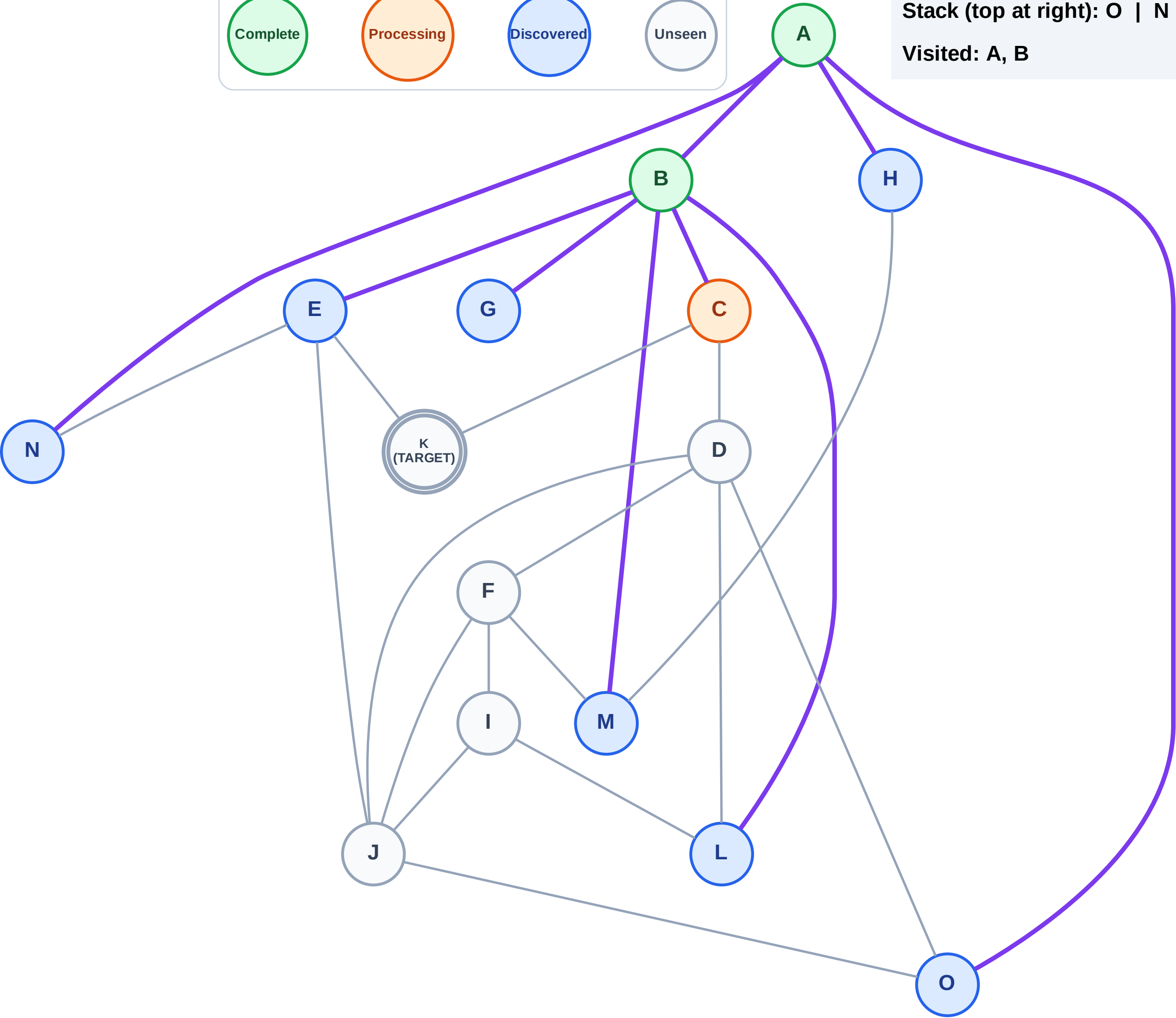
Processing

Discovered

Unseen

Stack (top at right): O | N | H | M | L | G | E

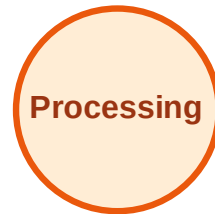
Visited: A, B



DFS Traversal

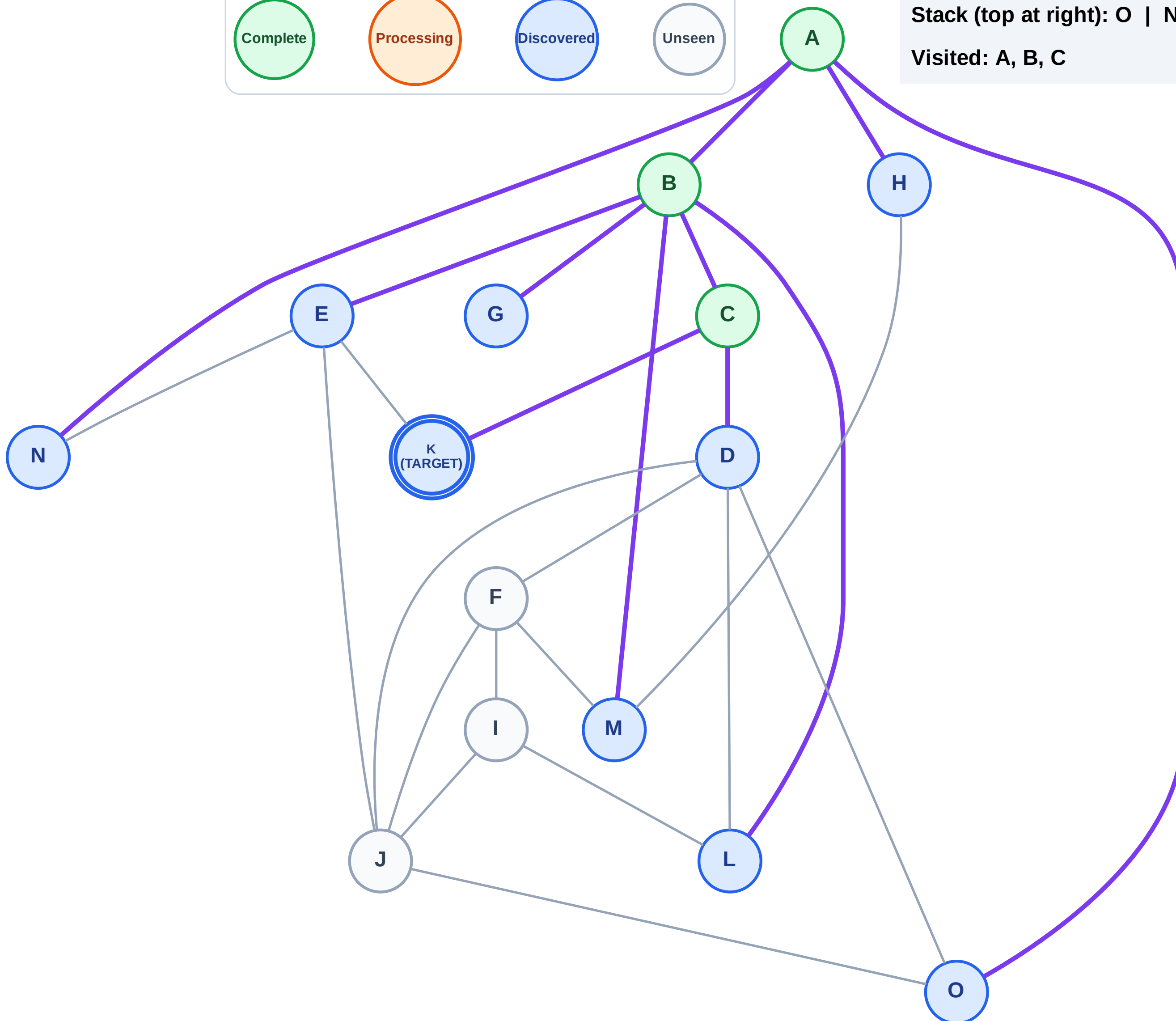
Step 7: Discover K, D from C

Legend



Stack (top at right): O | N | H | M | L | G | E | K | D

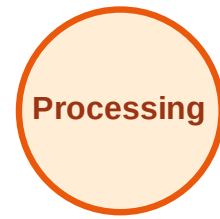
Visited: A, B, C



DFS Traversal

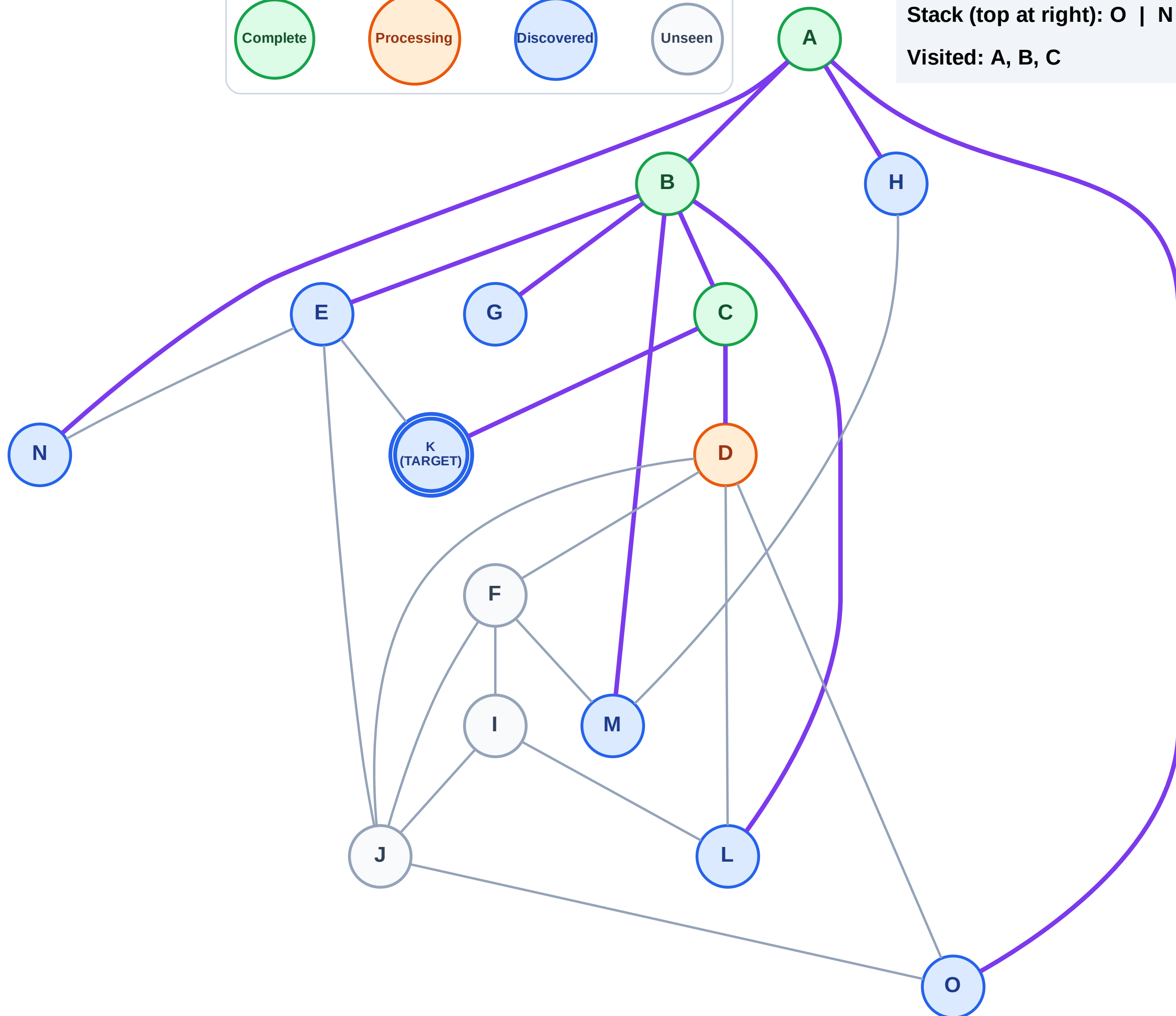
Step 8: Process node D

Legend



Stack (top at right): O | N | H | M | L | G | E | K

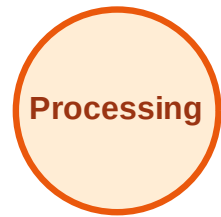
Visited: A, B, C



DFS Traversal

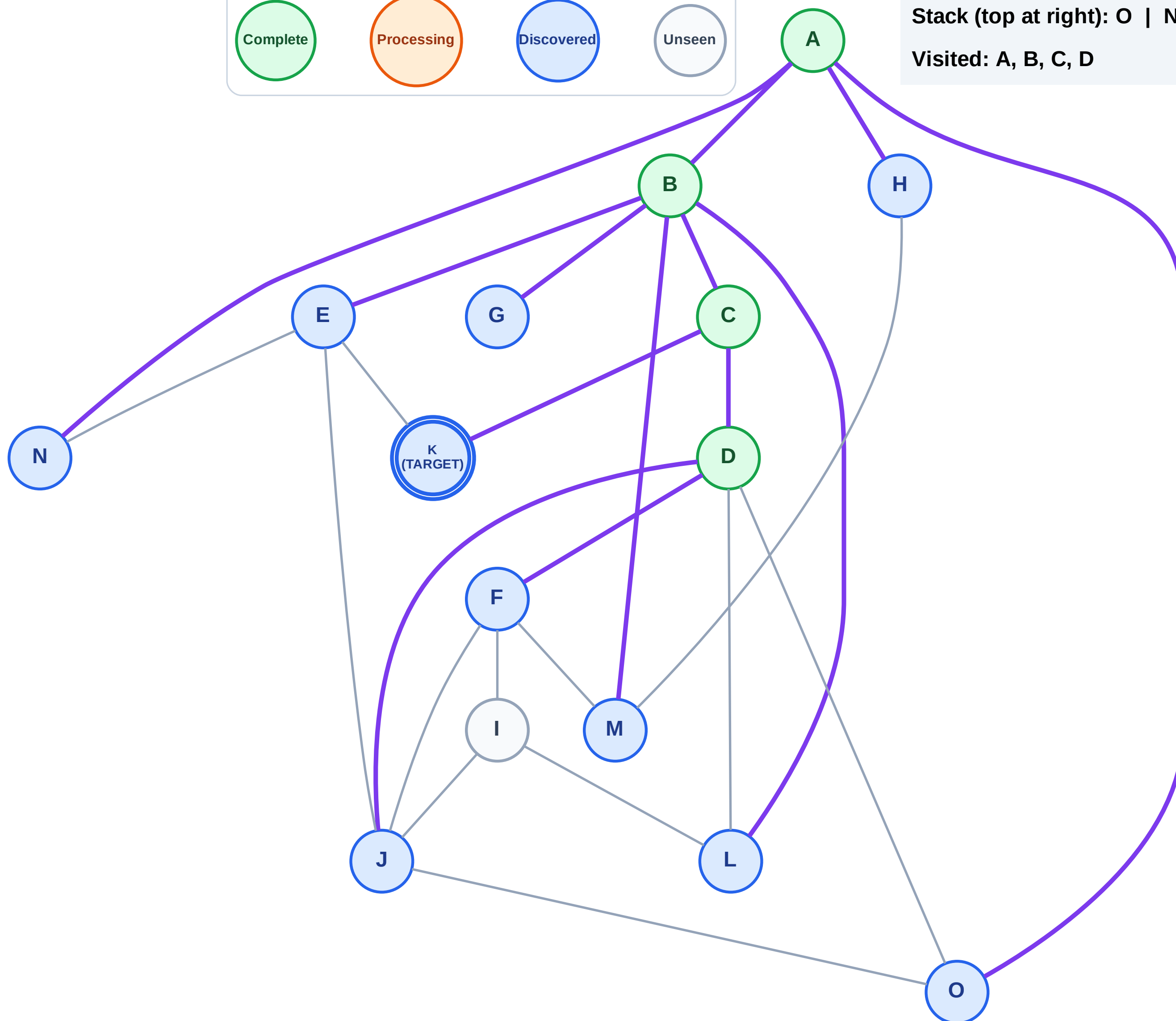
Step 9: Discover J, F from D

Legend



Stack (top at right): O | N | H | M | L | G | E | K | J | F

Visited: A, B, C, D



DFS Traversal

Step 10: Process node F

Legend

Complete

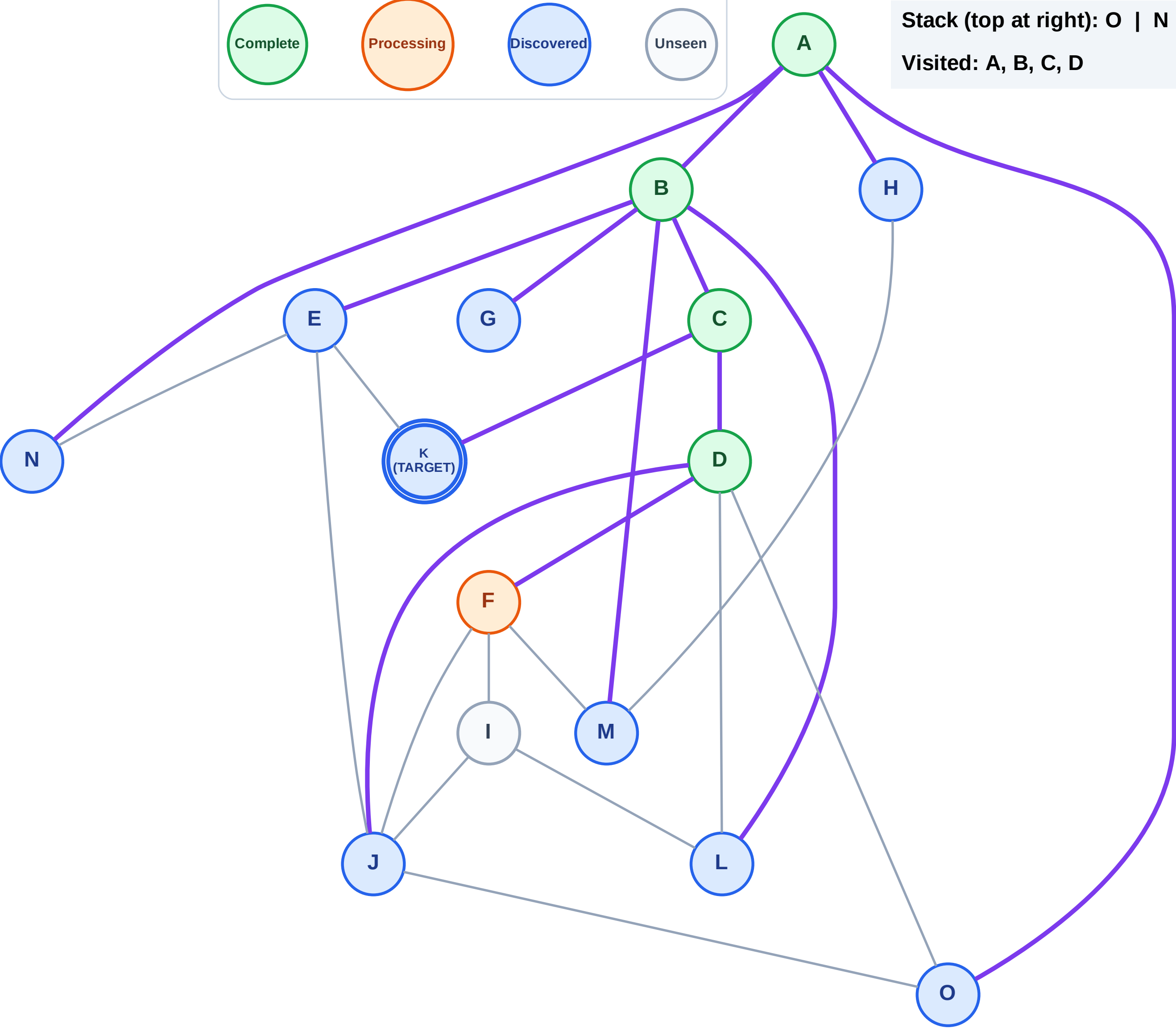
Processing

Discovered

Unseen

Stack (top at right): O | N | H | M | L | G | E | K | J

Visited: A, B, C, D



DFS Traversal

Step 11: Discover I from F

Legend

Complete

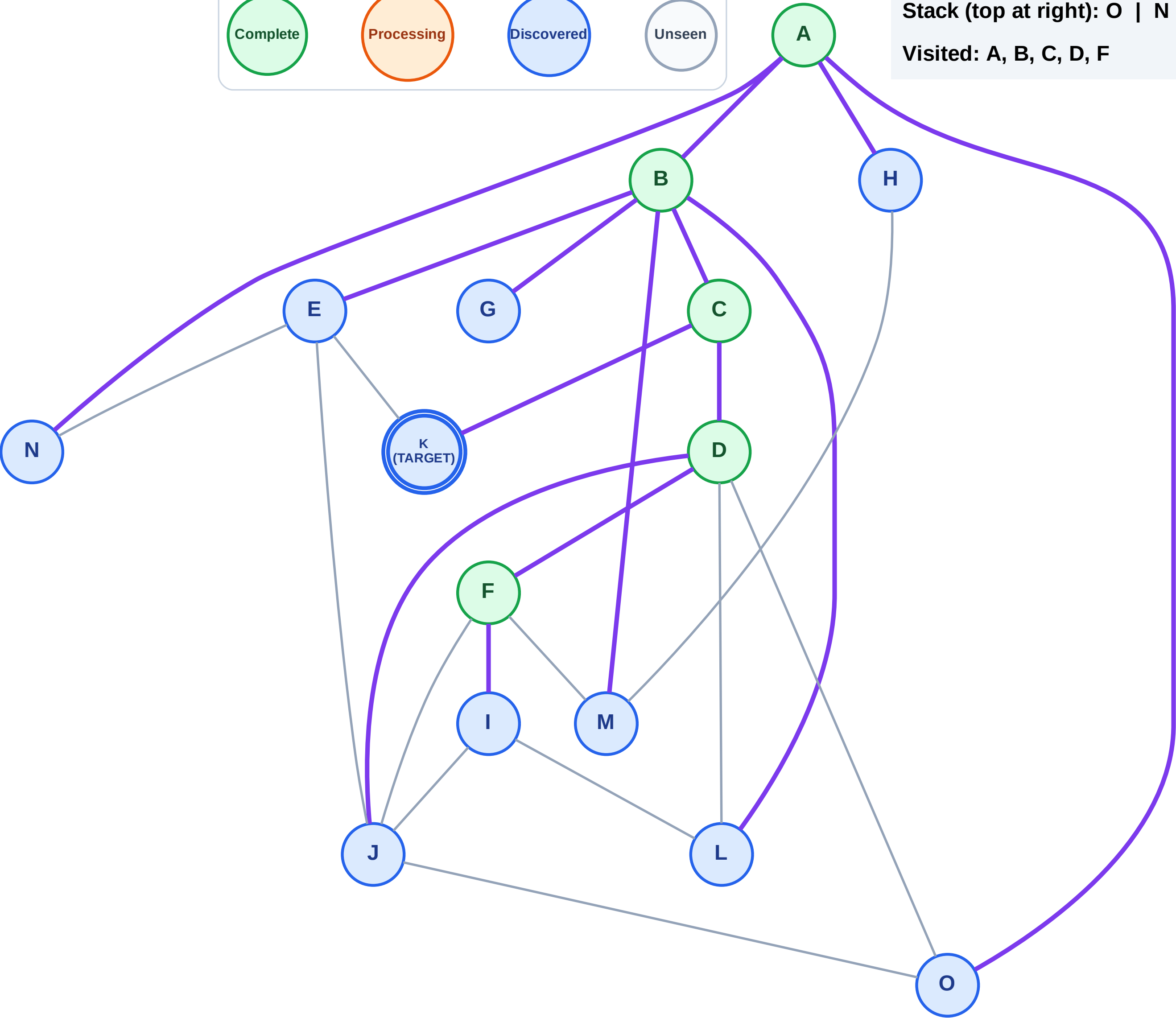
Processing

Discovered

Unseen

Stack (top at right): O | N | H | M | L | G | E | K | J | I

Visited: A, B, C, D, F



DFS Traversal

Step 12: Process node I

Legend

Complete

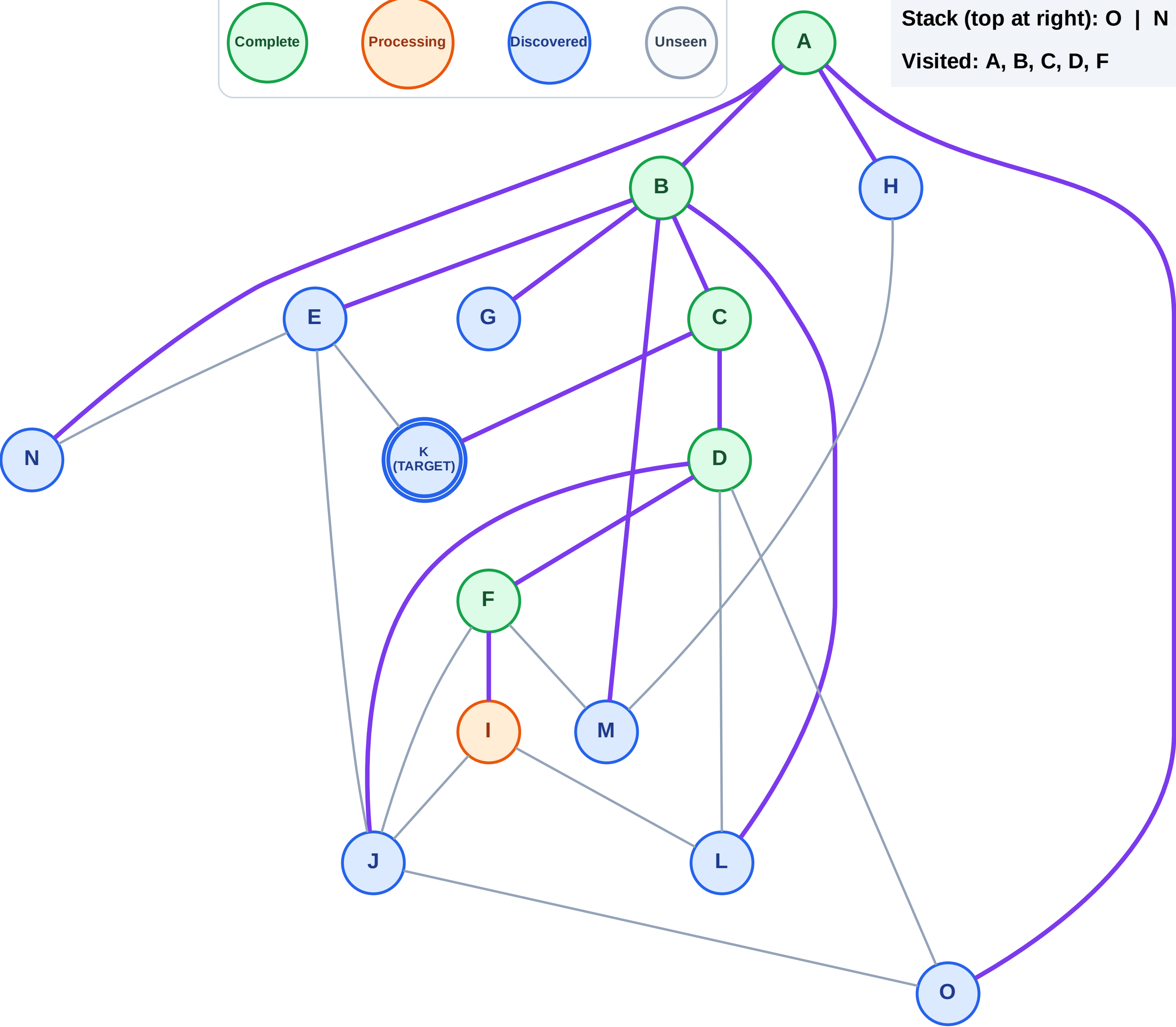
Processing

Discovered

Unseen

Stack (top at right): O | N | H | M | L | G | E | K | J

Visited: A, B, C, D, F



DFS Traversal

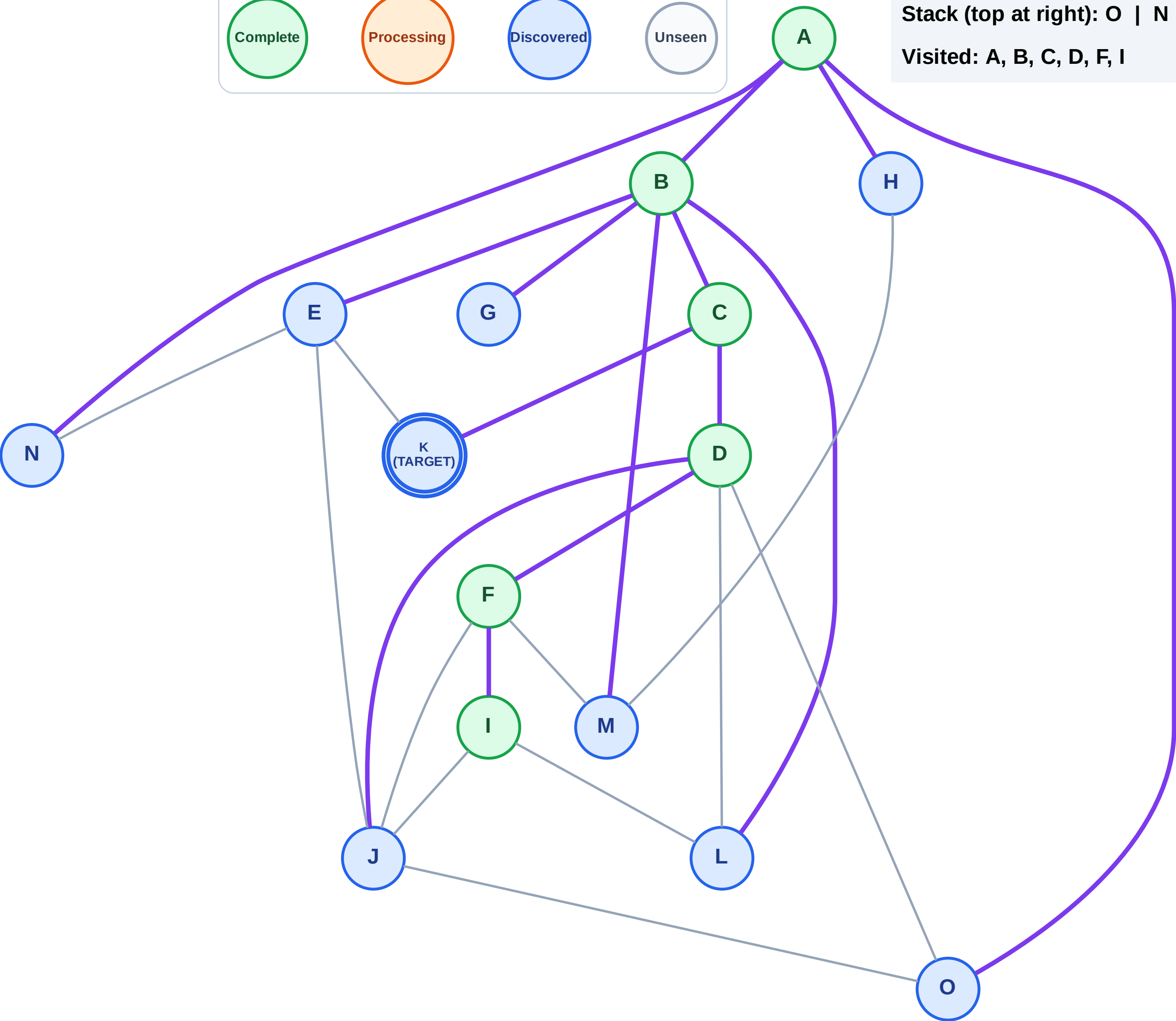
Step 13: No new neighbors from I

Legend



Stack (top at right): O | N | H | M | L | G | E | K | J

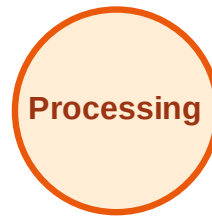
Visited: A, B, C, D, F, I



DFS Traversal

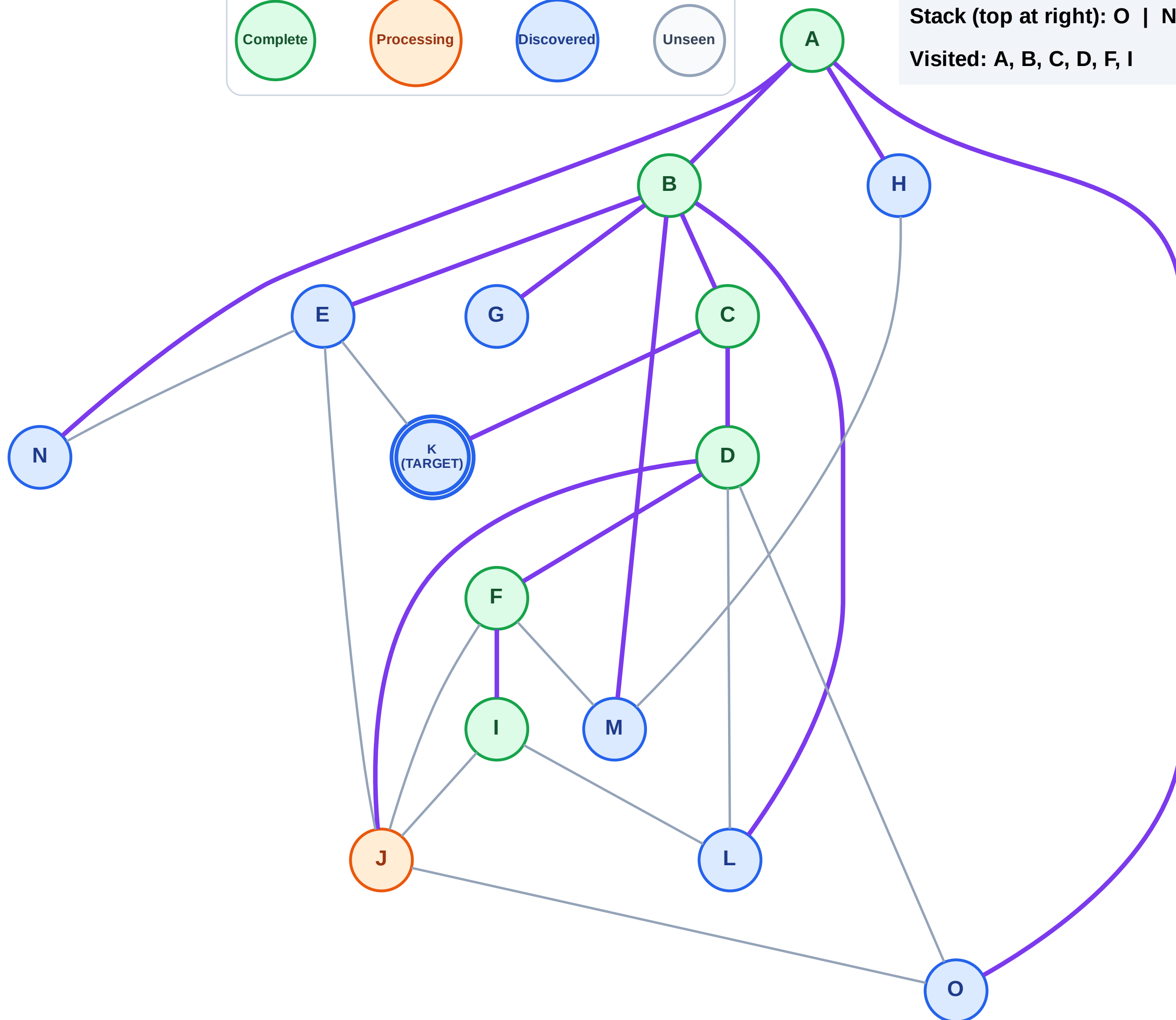
Step 14: Process node J

Legend



Stack (top at right): O | N | H | M | L | G | E | K

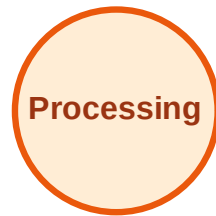
Visited: A, B, C, D, F, I



DFS Traversal

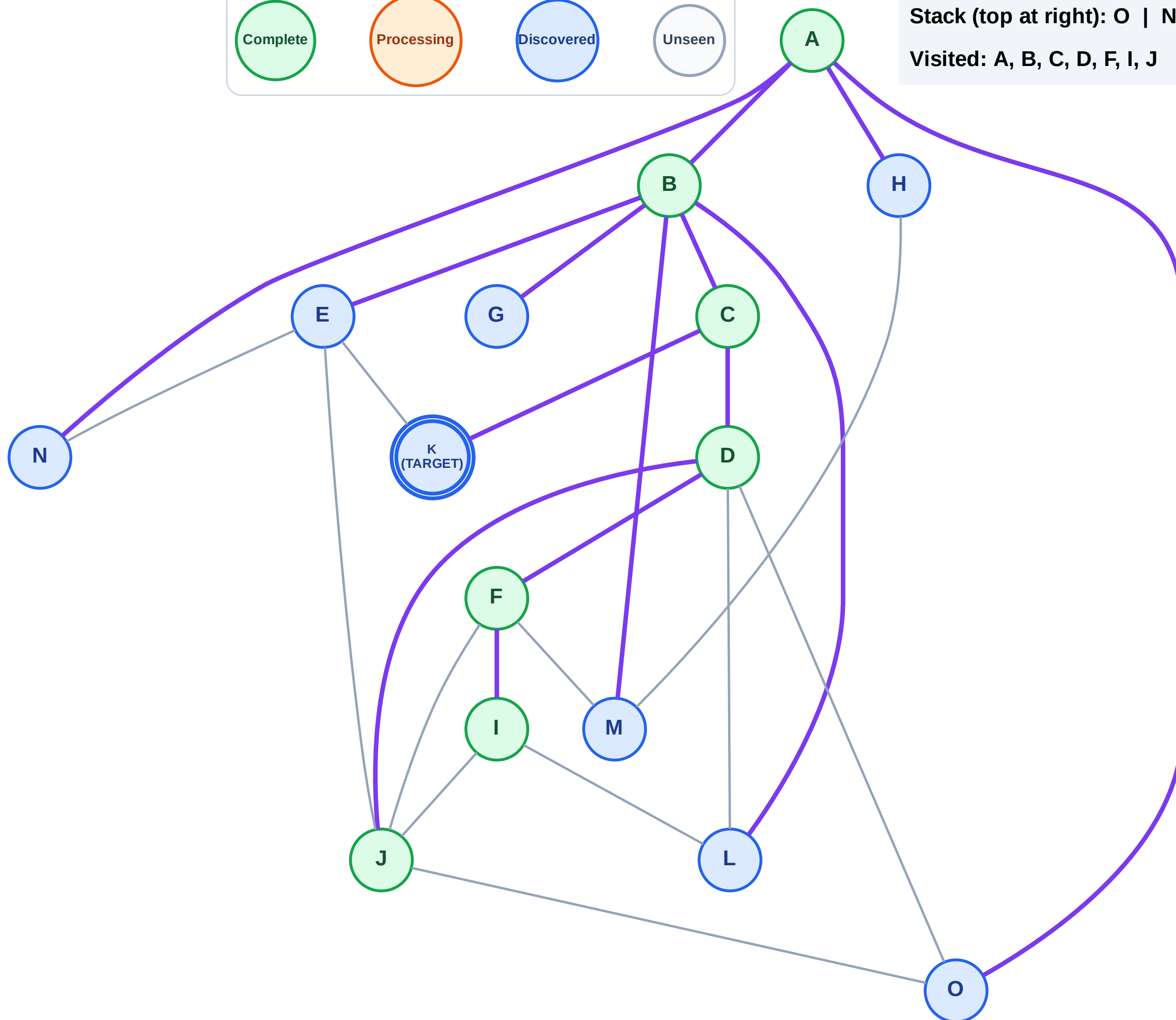
Step 15: No new neighbors from J

Legend



Stack (top at right): O | N | H | M | L | G | E | K

Visited: A, B, C, D, F, I, J



DFS Traversal

Step 16: Process node K

Legend

Complete

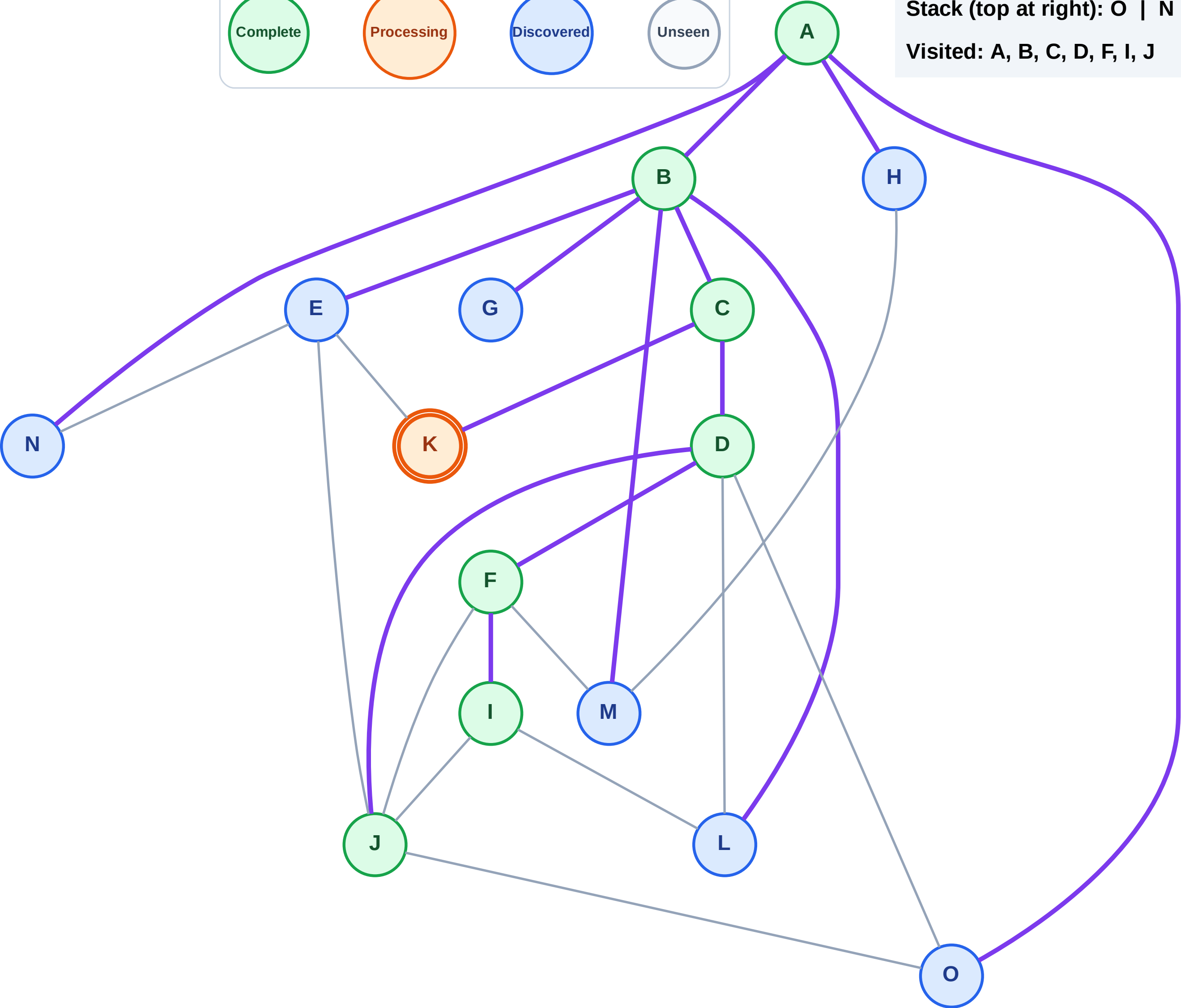
Processing

Discovered

Unseen

Stack (top at right): O | N | H | M | L | G | E

Visited: A, B, C, D, F, I, J



DFS Traversal

Step 17: TARGET NODE K REACHED! Path (3 hops): A → B → C → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | N | H | M | L | G | E

Visited: A, B, C, D, F, I, J, K

